

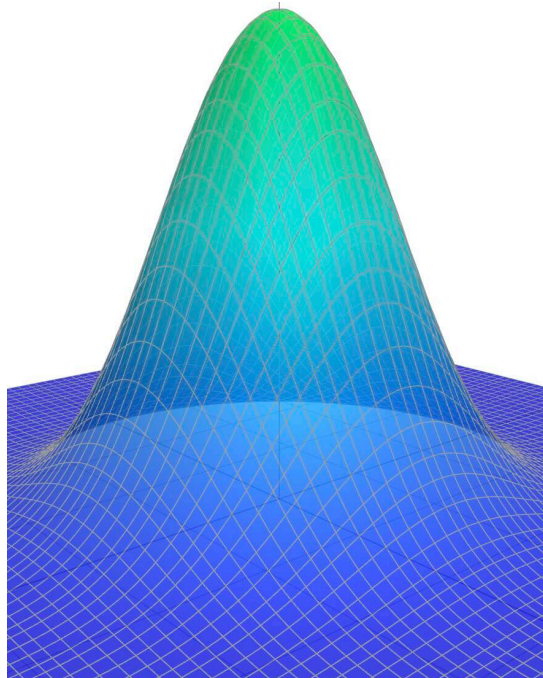
# Generalized Linear Models

Statistics III - CdL SSE

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# Homepage



- This unit will cover the following **topics**:
  - Exponential dispersion families
  - Likelihood, inference, and testing
  - Iteratively Re-weighted Least Squares (IRLS)
  - Deviance, model checking, and residuals
  - Model selection
- GLMs are regression models with a linear predictor, where the response variable follows an **exponential dispersion family**.
- The symbol  means that a few extra steps are discussed in the **handwritten notes**.

The content of this Unit is covered in **Chapter 2** of Salvan et al. (2020). Alternatively, see **Chapter 4** of Agresti (2015) or **Chapter 6** of Azzalini (2008).

# Introduction

# Preliminaries

- GLMs are a **class** of **regression models** in which a **response** random variable  $Y_i$  is modeled as a function of a vector of **covariates**  $\mathbf{x}_i \in \mathbb{R}^p$ .
- The random variables  $Y_i$  are not restricted to be Gaussian. For example:
  - $Y_i \in \{0, 1\}$ , known as **binary regression**
  - $Y_i \in \{0, 1, \dots\}$ , known as **count regression**
  - $Y_i \in (0, \infty)$  or  $Y_i \in (-\infty, \infty)$
- Gaussian linear models are a special case of GLMs, arising when  $Y_i \in (-\infty, \infty)$ .
- The **response random variables** are collected in the random vector  $\mathbf{Y} = (Y_1, \dots, Y_n)^T$ , whose **observed realization** is  $\mathbf{y} = (y_1, \dots, y_n)^T$ .
- The **design matrix**  $\mathbf{X}$  is an  $n \times p$  **non-stochastic** matrix containing the covariate values. The  $j$ th variable (column) is denoted by  $\tilde{\mathbf{x}}_j$ , while the  $i$ th observation (row) is  $\mathbf{x}_i$ .
- We assume that  $\mathbf{X}$  has **full rank**, that is,  $\text{rk}(\mathbf{X}) = p$  with  $p \leq n$ .

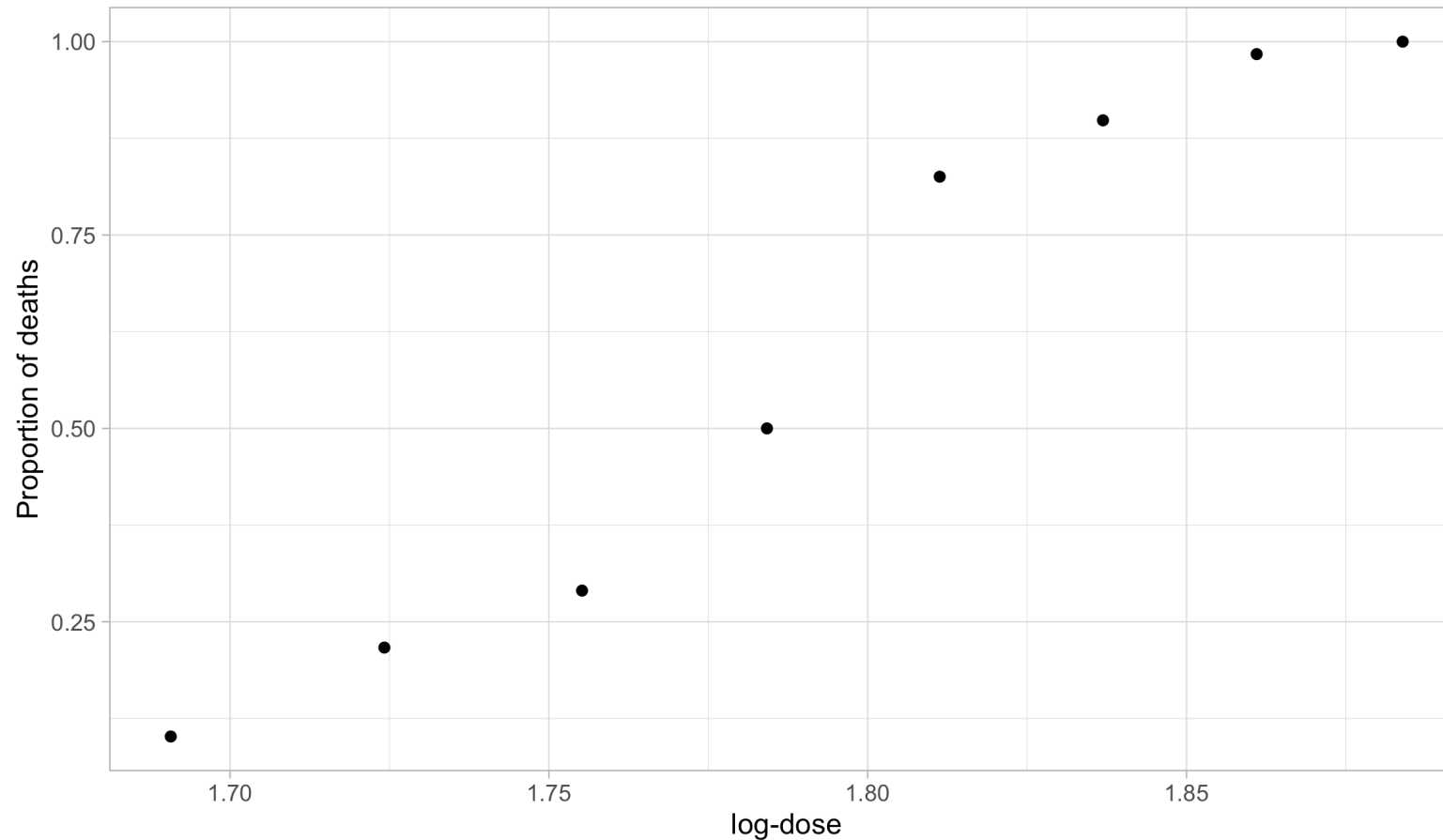
## Beetles data, from Bliss (1935)

- The **Beetles** dataset originates from Bliss (1935). It records the number of adult flour beetles that died after a 5-hour exposure to gaseous carbon disulphide.

| m  | deaths | logdose |
|----|--------|---------|
| 59 | 6      | 1.6907  |
| 60 | 13     | 1.7242  |
| 62 | 18     | 1.7552  |
| 56 | 28     | 1.7842  |
| 63 | 52     | 1.8113  |
| 59 | 53     | 1.8369  |
| 62 | 61     | 1.8610  |
| 60 | 60     | 1.8839  |

- We aim to predict the proportion of **deaths** as a function of **logdose**.
- Modeling death proportions directly with **linear models** is **inappropriate**. A **variable transformation** provides a more **principled** solution, but it comes with **drawbacks**.

## Beetles data, a dose-response plot



- There is a clear **positive** and **non-linear** relationship between the **proportion of deaths** and the logdose. The response variable takes values in  $[0, 1]$ .

# Modelling the Beetles data

- Let  $S_i$  be the number of dead beetles out of  $m_i$ , and let  $x_i$  denote the log-dose. By definition,  $S_i \in \{0, 1, \dots, m_i\}$  for  $i = 1, \dots, 8$ .
- It is natural to model each  $S_i$  as **independent binomial** random variables, counting the number of deaths out of  $m_i$  individuals. In other words:

$$S_i \stackrel{\text{ind}}{\sim} \text{Binomial}(m_i, \pi_i), \quad i = 1, \dots, 8,$$

where  $\pi_i$  is the **probability** of death at a given dose  $x_i$ . Moreover, let  $Y_i = S_i/m_i$  be the proportion of deaths, then:

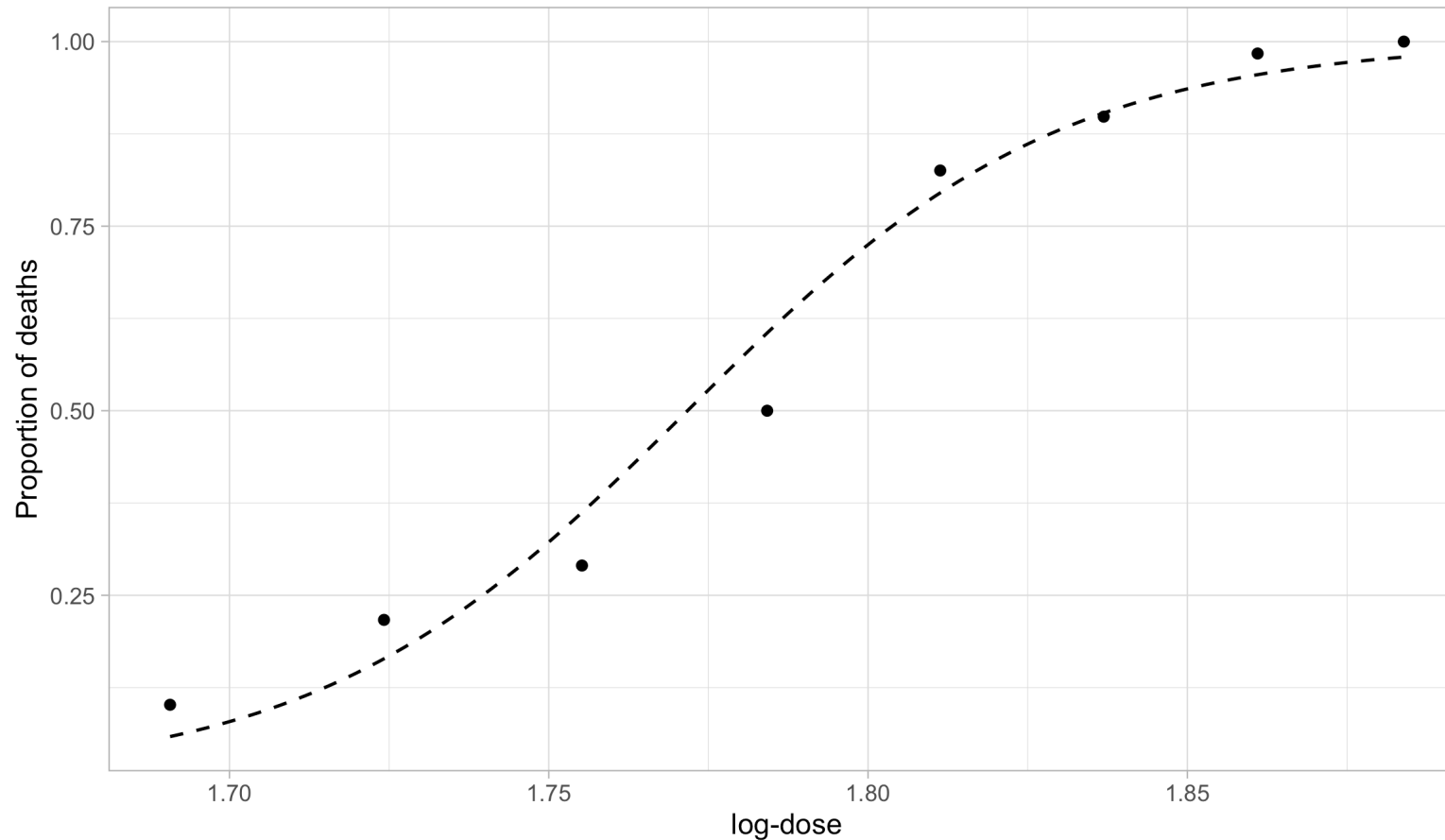
$$\mathbb{E}(Y_i) = \mathbb{E}\left(\frac{S_i}{m_i}\right) = \pi_i = \mu_i.$$

- A modeling approach, called **logistic regression**, specifies:

$$g(\pi_i) = \log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_1 + \beta_2 x_i \quad \implies \quad \pi_i = g^{-1}(\beta_1 + \beta_2 x_i) = \frac{\exp(\beta_1 + \beta_2 x_i)}{1 + \exp(\beta_1 + \beta_2 x_i)}.$$

for some parameters  $\beta_1, \beta_2 \in \mathbb{R}$ . **Note** that  $\pi_i \in (0, 1)$  by construction.

## Beetles data, fitted model



- The **maximum likelihood** estimates are  $\hat{\beta}_1 = -60.72$  and  $\hat{\beta}_2 = 34.3$ . This yields the **predictive curve**  $\hat{\pi}(x) = g^{-1}(\hat{\beta}_1 + \hat{\beta}_2 x)$ , which estimates the mean proportion  $\mathbb{E}(S_i/m_i)$ .

## A comparison with old tools I

Let  $Y_i = S_i/m_i$  be the proportion of deaths. A direct application of linear models implies:

$$Y_i = \beta_1 + \beta_2 x_i + \epsilon_i.$$

The coefficients  $\beta_1$  and  $\beta_2$  are then estimated using OLS using  $Y_i$  as response.

- The prediction  $\hat{\beta}_1 + \hat{\beta}_2 x_i$  is **unrestricted**, meaning it could produce values like “1.3” or “-2” as estimated **proportions**, which is clearly undesirable.
- The **additive structure**  $Y_i = \beta_1 + \beta_2 x_i + \epsilon_i$  cannot hold with **iid** errors  $\epsilon_i$ , because  $S_i$ , and thus  $Y_i$ , are **discrete**. As a result, the errors are always **heteroscedastic**.
- If  $m_i = 1$ , i.e. when the data are **binary**, all the above issues are **exacerbated**.

This approach is sometimes called the **linear probability model**. Before GLMs, it was considered acceptable despite its issues, but by modern standards it should **not be used**.

## A comparison with old tools II

We consider the **empirical logit** variable transformation of  $Y_i = S_i/m_i$ , obtaining

$$\text{logit}(\tilde{Y}_i) = \log\left(\frac{S_i + 0.5}{m_i - S_i + 0.5}\right) = \beta_1 + \beta_2 x_i + \epsilon_i, \quad \tilde{Y}_i = \frac{S_i + 0.5}{m_i + 1}.$$

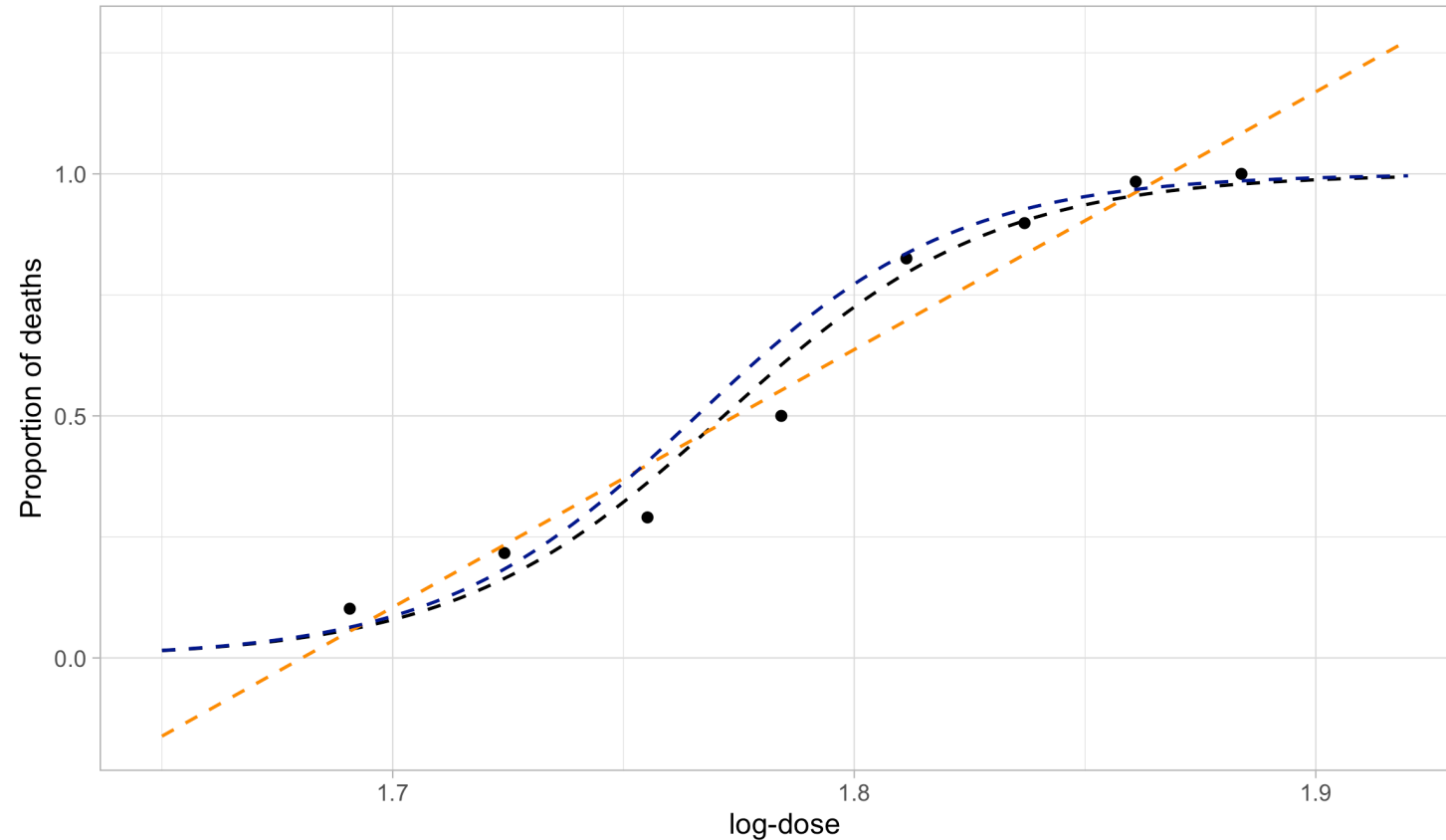
A correction term is necessary because otherwise  $g(\cdot) = \text{logit}(\cdot)$  is undefined. The predictions belong to  $(0, 1)$ , since

$$\hat{\pi}_i = g^{-1}(\hat{\beta}_1 + \hat{\beta}_2 x_i) = \frac{\exp(\hat{\beta}_1 + \hat{\beta}_2 x_i)}{1 + \exp(\hat{\beta}_1 + \hat{\beta}_2 x_i)},$$

in which  $\hat{\beta}_1$  and  $\hat{\beta}_2$  are estimated with OLS using  $\text{logit}(\tilde{Y}_i)$  as response.

- The **interpretation** of  $\hat{\beta}$  is less clear, as they refer to the mean of  $\text{logit}(\tilde{Y}_i)$  instead of  $\mathbb{E}(Y_i)$ .
- An arbitrary **boundary correction** is needed.
- Inference is problematic and requires further corrections, because of **heteroscedastic** errors.
- This approach is **not compatible** with the reasonable assumption  $S_i \sim \text{Binomial}(m_i, \pi_i)$ .

## A comparison with old tools III



- The black line is the predicted curve of a **logistic regression GLM**. The orange line is the predicted curve of a **linear model**. The blue line is the predictive curve of a **linear model** after an **empirical logit variable transformation**.

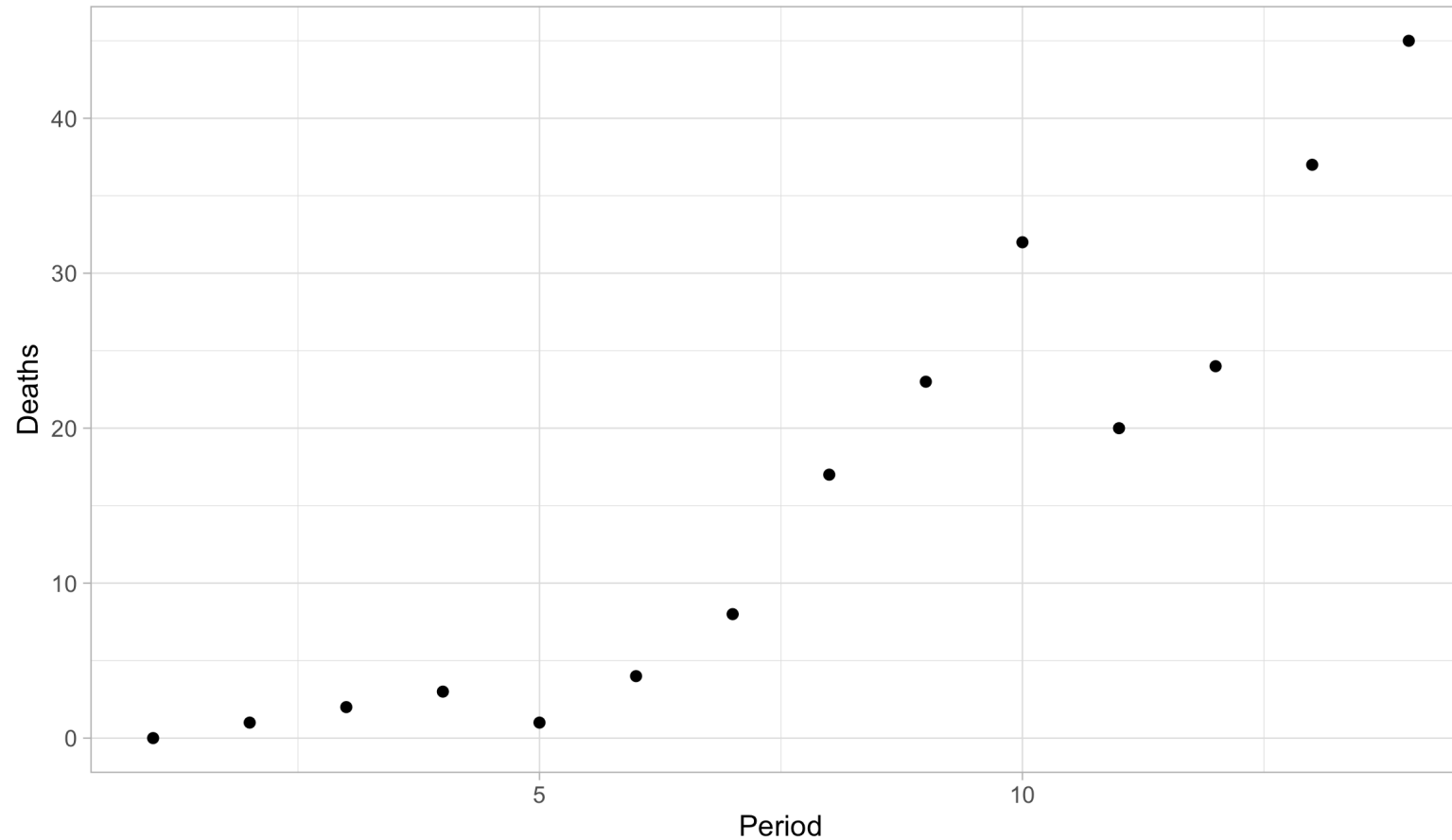
## Aids data

- Number of AIDS **deaths** in Australia in a sequence of three-month periods between 1983 and 1986.

|        | 1983-1 | 1983-2 | 1983-3 | 1983-4 | 1984-1 | 1984-2 | 1984-3 |
|--------|--------|--------|--------|--------|--------|--------|--------|
| deaths | 0      | 1      | 2      | 3      | 1      | 4      | 8      |
| period | 1      | 2      | 3      | 4      | 5      | 6      | 7      |
|        | 1984-4 | 1985-1 | 1985-2 | 1985-3 | 1985-4 | 1986-1 | 1986-2 |
| deaths | 17     | 23     | 32     | 20     | 24     | 37     | 45     |
| period | 8      | 9      | 10     | 11     | 12     | 13     | 14     |

- We are interested in predicting the number of **deaths** as a function of the **period** of time.
- The response variable  $Y_i \in \{0, 1, \dots\}$  is a non-negative **count**.

## Aids data, scatter plot



- There is a clear positive association between period and deaths. However, the increase appears to be **faster** than **linear**. Note that both the mean and the **variability** of  $Y_i$  increase over time.

# Modelling the **Aids** data

- Let  $Y_i$  be the number of deaths, and let  $x_i$  denote the period. By definition,  $Y_i \in \{0, 1, \dots\}$  are non-negative counts, for  $i = 1, \dots, 14$ .
- We model  $Y_i$  as **independent Poisson** random variables, counting the number of deaths:

$$Y_i \stackrel{\text{ind}}{\sim} \text{Poisson}(\mu_i), \quad i = 1, \dots, 14,$$

where  $\mu_i$  is the **mean** of  $Y_i$ , namely  $\mathbb{E}(Y_i) = \mu_i$ .

- A modeling approach, called **Poisson regression**, specifies:

$$g(\mu_i) = \log(\mu_i) = \beta_1 + \beta_2 x_i \quad \implies \quad \mu_i = g^{-1}(\beta_1 + \beta_2 x_i) = \exp(\beta_1 + \beta_2 x_i),$$

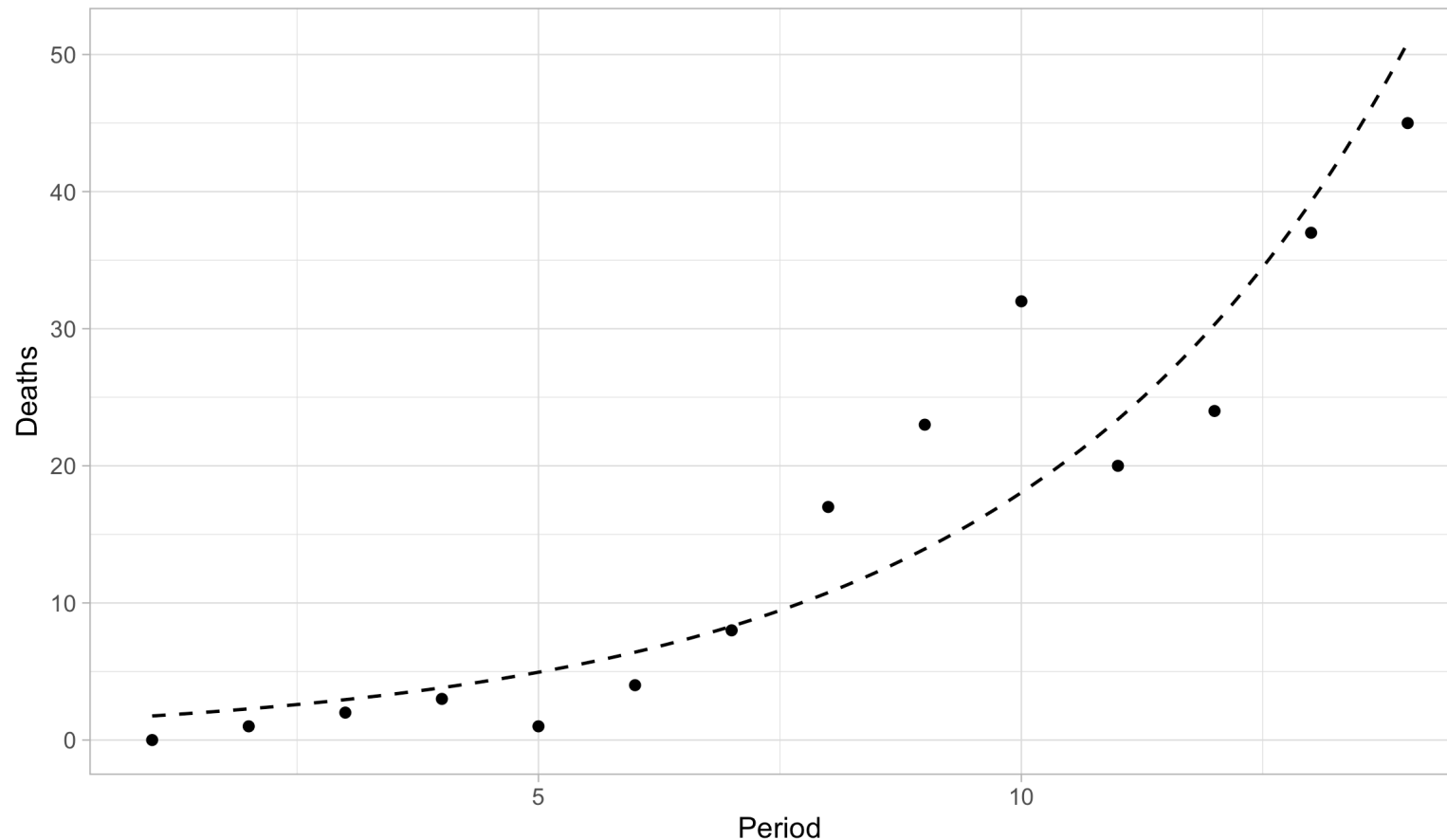
for some parameters  $\beta_1, \beta_2 \in \mathbb{R}$ . **Note** that  $\mu_i > 0$  by construction.

- Under this specification, the **variances** of the observations are

$$\text{var}(Y_i) = \mu_i = \exp(\beta_1 + \beta_2 x_i),$$

which increases with  $x$ , as desired. This implies that  $Y_1, \dots, Y_n$  are **heteroscedastic**, but this is not an issue in GLMs, as this aspect is **automatically accounted** for.

## Aids data, fitted model



- The **maximum likelihood** estimates are  $\hat{\beta}_1 = 0.304$  and  $\hat{\beta}_2 = 0.259$ . This yields the **predictive curve**  $\hat{\mu}(x) = \exp(\hat{\beta}_1 + \hat{\beta}_2 x)$ , which estimates the mean  $\mathbb{E}(Y_i)$ .

## A comparison with old tools I

We consider the **variance-stabilizing** transformation  $S_i = \sqrt{Y_i}$ , obtaining

$$\sqrt{Y_i} = \beta_1 + \beta_2 x_i + \epsilon_i.$$

The predictions belong to  $[0, \infty)$ , since

$$\hat{\mu}_i = (\hat{\beta}_1 + \hat{\beta}_2 x_i)^2,$$

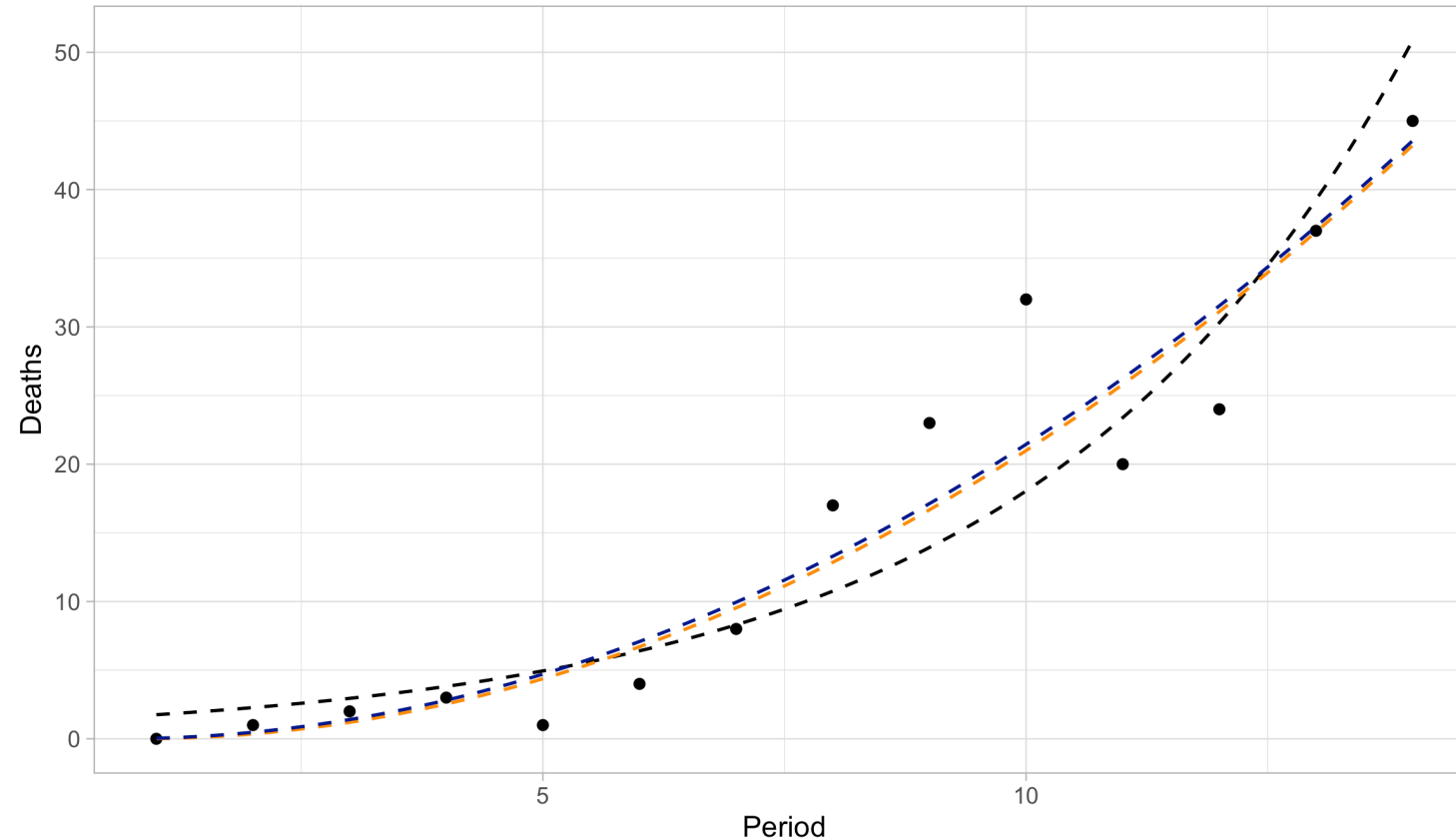
in which  $\hat{\beta}_1$  and  $\hat{\beta}_2$  are estimated with OLS using  $\sqrt{Y_i}$  as response.

- The **interpretation** of  $\hat{\beta}$  is less clear, as they refer to the mean of  $\sqrt{Y_i}$  instead of  $\mathbb{E}(Y_i)$ .
- This approach is **not compatible** with the reasonable assumption  $Y_i \sim \text{Poisson}(\mu_i)$  and it is only valid as an **asymptotic approximation**.

To compare such a model with a similar specification, we also fit another Poisson GLM in which

$$Y_i \stackrel{\text{ind}}{\sim} \text{Poisson}(\mu_i), \quad \sqrt{\mu_i} = \beta_1 + \beta_2 x_i, \quad i = 1, \dots, 14.$$

## A comparison with old tools II



- The black line is the predicted curve of a **Poisson regression GLM** with logarithmic link. The orange line is the predicted curve of a **linear model** with a **square-root transformation**. The blue line is the predictive curve of a **Poisson regression GLM** with **square-root link**.

# The components of a GLM

- **Random component.** This specifies the probability distribution of the response variable  $Y_i$ . The observations  $\mathbf{y} = (y_1, \dots, y_n)$  on that distribution are treated as **independent**.
- **Linear predictor.** For a parameter vector  $\beta = (\beta_1, \dots, \beta_p)^T$  and an  $n \times p$  design matrix  $\mathbf{X}$ , the linear predictor is  $\eta = \mathbf{X}\beta$ . We will also write

$$\eta_i = \mathbf{x}_i^T \beta = x_{i1}\beta_1 + \dots + x_{ip}\beta_p, \quad i = 1, \dots, n.$$

- **Link function.** This is an invertible and differentiable function  $g(\cdot)$  applied to each component of the mean  $\mu_i = \mathbb{E}(Y_i)$  that relates it to the linear predictor:

$$g(\mu_i) = \eta_i = \mathbf{x}_i^T \beta, \quad \implies \quad \mu_i = g^{-1}(\eta_i) = g^{-1}(\mathbf{x}_i^T \beta).$$

Note that, in general, we **cannot** express the response in an additive way  $Y_i = g^{-1}(\eta_i) + \epsilon_i$ .

# Random component of a GLM

- In GLMs the random variables  $Y_i$  are **independent** and they are distributed according to an **exponential dispersion family**, whose definition will be provided in a few slides.
- The **distributions most commonly** used in Statistics, such as the normal, binomial, gamma, and Poisson, are exponential family distributions.
- Exponential dispersion families are **characterized** by their **mean** and **variance**. Let  $v(\mu) > 0$  be a function of the mean, called **variance function** and let  $a_i(\phi) > 0$  be functions of an additional unknown parameter  $\phi > 0$  called **dispersion**.

In a GLM the observations are independent draws from a distribution  $\text{ED}(\mu_i, a_i(\phi)v(\mu_i))$ :

$$Y_i \stackrel{\text{ind}}{\sim} \text{ED}(\mu_i, a_i(\phi)v(\mu_i)), \quad \mathbb{E}(Y_i) = \mu_i, \quad g(\mu_i) = \mathbf{x}_i^T \boldsymbol{\beta},$$

with  $\mu_i \in \mathcal{M}$ . Moreover, the **variance** is connected to the **mean** via  $v(\mu)$ :

$$\text{var}(Y_i) = a_i(\phi)v(\mu_i),$$

where  $a_i(\phi) = \phi/\omega_i$  and  $\omega_i$  are **known weights**. Special cases are  $a_i(\phi) = \phi$  and  $a_i(\phi) = 1$ .

## Notable examples

In a **Gaussian linear model** we consider the **identity link**  $g(\mu) = \mu$  and let

$$Y_i \stackrel{\text{ind}}{\sim} \text{N}(\mu_i, \sigma^2), \quad \mu_i = \mathbf{x}_i^T \beta.$$

The unknown variance  $\sigma^2 = \phi$  is called **dispersion** in GLMs. The **parameter space** is  $\mathcal{M} = \mathbb{R}$ , whereas  $a_i(\phi) = \phi$  and the variance function is **constant**  $v(\mu) = 1$  (homoscedasticity).

In a **binomial regression model** with **logit link**  $g(\mu) = \text{logit}(\mu)$  we let  $Y_i = S_i/m_i$  and

$$S_i \stackrel{\text{ind}}{\sim} \text{Binomial}(m_i, \pi_i), \quad \mathbb{E}(Y_i) = \pi_i = \mu_i, \quad \text{logit}(\mu_i) = \mathbf{x}_i^T \beta.$$

We have  $a_i(\phi) = 1/m_i$  and  $v(\mu) = \mu(1 - \mu)$ . There is **no dispersion** parameter.

In **Poisson regression** with **logarithmic link**  $g(\mu) = \log(\mu)$  we let

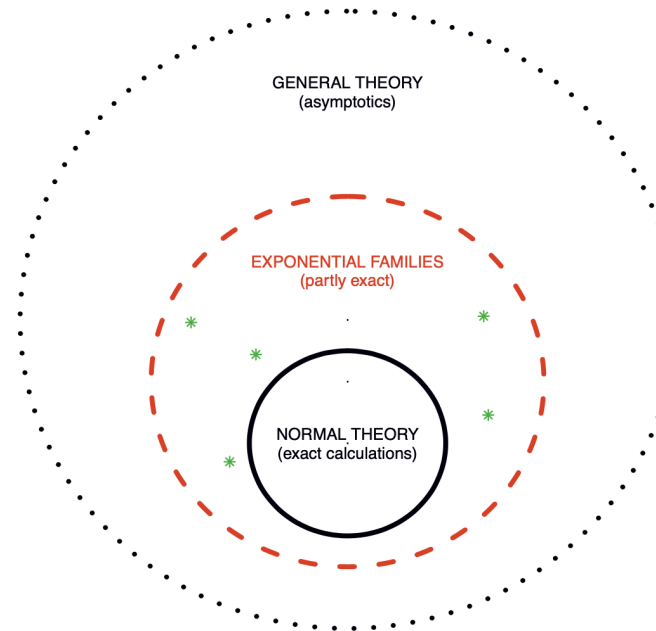
$$Y_i \stackrel{\text{ind}}{\sim} \text{Poisson}(\mu_i), \quad \log(\mu_i) = \mathbf{x}_i^T \beta.$$

We have  $a_i(\phi) = 1$  and  $v(\mu) = \mu$ . There is **no dispersion** parameter.

# Exponential dispersion families

# Overview

- Figure 1 of Efron (2023). **Three levels** of statistical modeling.



- The **prime role** of **exponential families** in the theory of statistical inference was first emphasized by Fisher (1934).
- Most **well-known distributions**—such as Gaussian, Poisson, Binomial, and Gamma—are instances of exponential families.

## Exponential dispersion family: definition

The **density** of  $Y_i$  belongs to an **exponential dispersion family** if it can be written as

$$p(y_i; \theta_i, \phi) = \exp \left\{ \frac{\theta_i y_i - b(\theta_i)}{a_i(\phi)} + c(y_i, \phi) \right\},$$

where  $y_i \in \mathcal{Y} \subseteq \mathbb{R}$ ,  $\theta_i \in \Theta \subseteq \mathbb{R}$  and  $a_i(\phi) = \phi/\omega_i$  where  $\omega_i$  are **known positive weights**. The parameter  $\theta_i$  is called **natural parameter** while  $\phi$  is called **dispersion** parameter.

- By specifying the functions  $a_i(\cdot)$ ,  $b(\cdot)$  and  $c(\cdot)$  one obtains a particular **parametric model**.
- The support  $\mathcal{Y}$  of  $Y_i$  does not depend on the parameters  $\phi$  or  $\theta_i$  and  $b(\cdot)$  can be **differentiated** infinitely many times. In particular, this is a **regular statistical model**.
- As mentioned, special cases are  $a_i(\phi) = \phi$  and  $a_i(\phi) = 1$ . When  $a_i(\phi) = 1$  and  $c(y_i, \phi) = c(y_i)$  we obtain

$$p(y_i; \theta_i) = \exp \{ \theta_i y_i - b(\theta_i) + c(y_i) \},$$

which is called **natural exponential family** of order 1.

# Mean and variance I

- Let us consider the **log-likelihood** contribution of the  $i$ th observations, which is defined as

$$\ell(\theta_i, \phi; y_i) = \log p(y_i; \theta_i, \phi) = \frac{\theta_i y_i - b(\theta_i)}{a_i(\phi)} + c(y_i, \phi).$$

If you prefer, this is the log-likelihood when the sample size  $n = 1$  and we only observe  $Y_i$ .

- The **score** and **Hessian** functions, namely the first and second derivatives with respect to  $\theta_i$ , are

$$\frac{\partial}{\partial \theta_i} \ell(\theta_i, \phi; y_i) = \frac{y_i - b'(\theta_i)}{a_i(\phi)}, \quad \frac{\partial^2}{\partial \theta_i^2} \ell(\theta_i, \phi; y_i) = \frac{-b''(\theta_i)}{a_i(\phi)}.$$

where  $b'(\cdot)$  and  $b''(\cdot)$  denote the **first** and **second** derivative of  $b(\cdot)$ .

- Recall the following **Bartlett identities**, valid in any **regular** statistical model:

$$\mathbb{E} \left( \frac{\partial}{\partial \theta_i} \ell(\theta_i, \phi; Y_i) \right) = 0,$$

$$\mathbb{E} \left\{ \left( \frac{\partial}{\partial \theta_i} \ell(\theta_i, \phi; Y_i) \right)^2 \right\} = \text{var} \left( \frac{\partial}{\partial \theta_i} \ell(\theta_i, \phi; Y_i) \right) = \mathbb{E} \left( -\frac{\partial^2}{\partial \theta_i^2} \ell(\theta_i, \phi; Y_i) \right).$$

## Mean and variance II

- Specializing Bartlett identities in **exponential dispersion families**, we obtain

$$\mathbb{E} \left( \frac{Y_i - b'(\theta_i)}{a_i(\phi)} \right) = 0, \quad \text{var} \left( \frac{Y_i - b'(\theta_i)}{a_i(\phi)} \right) = \frac{\text{var}(Y_i)}{a_i(\phi)^2} = \frac{b''(\theta_i)}{a_i(\phi)}.$$

Re-arranging the terms, we finally get the following key result.

Let  $Y_i$  be an exponential dispersion family, identified by the functions  $a_i(\cdot)$ ,  $b(\cdot)$  and  $c(\cdot)$ , and with natural parameter  $\theta_i$ . Then the **mean** and the **variance** of  $Y_i$  equal

$$\mathbb{E}(Y_i) = b'(\theta_i), \quad \text{var}(Y_i) = a_i(\phi)b''(\theta_i).$$

- The mean  $\mu_i = b'(\theta_i)$  does **not** depend on the **dispersion** parameter.
- We have  $b''(\cdot) > 0$  because  $\text{var}(Y_i) > 0$ , which means that  $b(\cdot)$  is a **convex function**.
- Moreover, the function  $b'(\theta)$  is **continuous** and **monotone increasing** and hence **invertible**.

The function  $b(\cdot)$  is related to the moment generating function of  $Y_i$ . Thus, higher order derivatives of  $b(\cdot)$  allow the calculation of skewness, kurtosis, etc.

# Mean parametrization, variance function

Let  $Y_i$  be an exponential dispersion family, identified by the functions  $a_i(\cdot)$ ,  $b(\cdot)$  and  $c(\cdot)$ , and with **natural parameter**  $\theta_i$ , then

$$\mu(\theta_i) := \mu_i = \mathbb{E}(Y_i) = b'(\theta_i).$$

The function  $\mu(\cdot) : \Theta \rightarrow \mathcal{M}$  is **one-to-one** and invertible, that is, a **reparametrization** of  $\theta_i$ . We call  $\mu_i$  the **mean parametrization** of an exponential dispersion family.

- The **inverse** relationship, re-obtaining  $\theta_i$  as a function of  $\mu_i$ , is denoted with

$$\theta_i = \theta(\mu_i) = b'^{-1}(\mu_i).$$

- Using this notation, we can express the variance of  $Y_i$  as a function of  $\mu_i$  as follows

$$\text{var}(Y_i) = a_i(\phi)b''(\theta_i) = a_i(\phi)b''(\theta(\mu_i)) = a_i(\phi)v(\mu_i),$$

where  $v(\mu_i) := b''(\theta(\mu_i))$  is the **variance function**.

- The domain  $\mathcal{M}$  and the variance function  $v(\mu)$  **characterize** the function  $b(\cdot)$  and the entire distribution, for any given  $a_i(\phi)$ . This justifies the **notation**  $Y_i \sim \text{ED}(\mu_i, a_i(\phi)v(\mu_i))$ .

# Gaussian distribution

- Let  $Y_i \sim N(\mu_i, \sigma^2)$ . The **density** function of  $Y_i$  can be written as

$$\begin{aligned} p(y_i; \mu_i, \sigma^2) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} (y_i - \mu_i)^2 \right\} \\ &= \exp \left\{ \frac{y_i \mu_i - \mu_i^2/2}{\sigma^2} - \frac{\log(2\pi\sigma^2)}{2} - \frac{y_i^2}{2\sigma^2} \right\} \end{aligned}$$

- Then, we can recognize the following relationships:

$$\theta_i = \theta(\mu_i) = \mu_i, \quad a_i(\phi) = \phi = \sigma^2, \quad b(\theta_i) = \frac{\theta_i^2}{2}, \quad c(y_i, \phi) = -\frac{\log(2\pi\phi)}{2} - \frac{y_i^2}{2\phi}.$$

In the Gaussian case, the **mean parametrization** and the **natural parametrization** coincide. Moreover, the **dispersion**  $\phi$  coincides with the **variance**  $\sigma^2$ .

- Using the results we previously discussed, we obtain the well-known relationships

$$\mathbb{E}(Y_i) = b'(\theta_i) = \theta_i, \quad \text{var}(Y_i) = a_i(\phi)b''(\theta_i) = \phi.$$

The **variance function**  $v(\mu_i) = 1$  is **constant**. We will write  $Y_i \sim \text{ED}(\mu_i, \phi)$  with  $\mu_i \in \mathcal{M} = \mathbb{R}$ .

# Poisson distribution

Let  $Y_i \sim \text{Poisson}(\mu_i)$ . The **pmf** of  $Y_i$  can be written as

$$\begin{aligned} p(y_i; \mu_i) &= \frac{\mu_i^{y_i} e^{-\mu_i}}{y_i!} = \exp\{y_i \log(\mu_i) - \mu_i - \log(y_i!)\} \\ &= \exp\{y_i \theta_i - e^{\theta_i} - \log(y_i!)\}, \quad y_i = 0, 1, 2, \dots \end{aligned}$$

- Then, we can recognize the following relationships:

$$\begin{aligned} \theta_i &= \theta(\mu_i) = \log(\mu_i), & a_i(\phi) &= 1, \\ b(\theta_i) &= e^{\theta_i}, & c(y_i, \phi) &= c(y_i) = -\log(y_i!). \end{aligned}$$

There is **no dispersion** parameter since  $a_i(\phi) = 1$ .

- Using the results we previously discussed, we obtain the well-known relationships

$$\begin{aligned} \mathbb{E}(Y_i) &= b'(\theta_i) = e^{\theta_i} = \mu_i, \\ \text{var}(Y_i) &= a_i(\phi) b''(\theta_i) = e^{\theta_i} = \mu_i. \end{aligned}$$

The **variance function**  $v(\mu_i) = \mu_i$  is **linear**. We will write  $Y_i \sim \text{ED}(\mu_i, \mu_i)$  with  $\mu_i \in (0, \infty)$ .

## Gamma distribution I

- Let  $Y_i \sim \text{Gamma}(\alpha, \lambda_i)$ . The **density** function of  $Y_i$  can be written as

$$\begin{aligned}
 p(y_i; \alpha, \lambda_i) &= \frac{\lambda_i^\alpha y_i^{\alpha-1} e^{-\lambda_i y_i}}{\Gamma(\alpha)} \\
 &= \exp \{ \alpha \log \lambda_i - \lambda_i y_i + (\alpha - 1) \log y_i - \log \Gamma(\alpha) \} \\
 &= \exp \left\{ \alpha \left( \log \lambda_i - \frac{\lambda_i}{\alpha} y_i \right) + (\alpha - 1) \log y_i - \log \Gamma(\alpha) \right\} \\
 &= \exp \left\{ \frac{\theta_i y_i + \log(-\theta_i)}{\phi} - (1/\phi) \log \phi + (1/\phi - 1) \log y_i - \log \Gamma(1/\phi) \right\}, \quad y_i > 0,
 \end{aligned}$$

having defined the **dispersion**  $\phi = 1/\alpha$  and the **natural parameter**  $\theta_i = -\lambda_i/\alpha$ .

- Then, we can recognize the following relationships:

$$\begin{aligned}
 a_i(\phi) &= \phi, & b(\theta_i) &= -\log(-\theta_i), \\
 c(y_i, \phi) &= -(1/\phi) \log \phi + (1/\phi - 1) \log y_i - \log \Gamma(1/\phi).
 \end{aligned}$$

## Gamma distribution II

- Using the results we previously discussed, we obtain the well-known relationships

$$\mathbb{E}(Y_i) = b'(\theta_i) = -\frac{1}{\theta_i} = \frac{\alpha}{\lambda_i} = \mu_i, \quad \text{var}(Y_i) = a_i(\phi)b''(\theta_i) = \frac{\phi}{\theta_i^2} = \frac{\alpha}{\lambda_i^2}.$$

- At the same time, we can write the **inverse** relationship linking  $\theta_i$  to the **mean** as

$$\theta_i = \theta(\mu_i) = -\frac{1}{\mu_i}$$

from which we finally obtain the following **quadratic** variance function

$$v(\mu_i) = \mu_i^2.$$

- We will write  $Y_i \sim \text{ED}(\mu_i, \phi\mu_i^2)$  with  $\mu_i \in (0, \infty)$ .

## Binomial distribution I

- Let  $S_i \sim \text{Binomial}(m_i, \pi_i)$ , with  $\pi_i \in (0, 1)$ . The random variable  $Y_i = S_i/m_i$  has **density**

$$\begin{aligned} p(y_i; m_i, \pi_i) &= \binom{m_i}{m_i y_i} \pi_i^{m_i y_i} (1 - \pi_i)^{m_i - m_i y_i} \\ &= \binom{m_i}{m_i y_i} \left( \frac{\pi_i}{1 - \pi_i} \right)^{m_i y_i} (1 - \pi_i)^{m_i} \\ &= \exp \left\{ m_i y_i \log \left( \frac{\pi_i}{1 - \pi_i} \right) + m_i \log(1 - \pi_i) + \log \binom{m_i}{m_i y_i} \right\}, \end{aligned}$$

for  $y_i \in \{0, 1/m_i, 2/m_i, \dots, 1\}$ . This can be written as

$$p(y_i; m_i, \pi_i) = \exp \left\{ \frac{y_i \theta_i - \log\{1 + \exp(\theta_i)\}}{1/m_i} + \log \binom{m_i}{m_i y_i} \right\},$$

where the **natural parameter** is  $\theta_i = \text{logit}(\pi_i) = \log\{\pi_i/(1 - \pi_i)\}$ .

## Binomial distribution II

- Note that  $\mathbb{E}(Y_i) = \mathbb{E}(S_i/m_i) = \pi_i = \mu_i$ . This means there is **no dispersion** parameter  $\phi$  and

$$\theta_i = \text{logit}(\mu_i), \quad a_i(\phi) = \frac{1}{m_i}, \quad b(\theta_i) = \log\{1 + \exp(\theta_i)\}, \quad c(y_i) = \log\left(\frac{m_i}{m_i y_i}\right).$$

- Using the general formulas therefore we obtain

$$\begin{aligned} \mathbb{E}(Y_i) &= b'(\theta_i) = \frac{\exp(\theta_i)}{1 + \exp(\theta_i)} = \mu_i, \\ \text{var}(Y_i) &= a_i(\phi)b''(\theta_i) = \frac{1}{m_i} \frac{\exp(\theta_i)}{[1 + \exp(\theta_i)]^2} = \frac{\mu_i(1 - \mu_i)}{m_i}, \end{aligned}$$

from which we obtain that the **variance function**  $v(\mu_i) = \mu_i(1 - \mu_i)$  is quadratic.

- We will write  $Y_i \sim \text{ED}(\mu_i, \mu_i(1 - \mu_i)/m_i)$  with  $\mu_i \in \mathcal{M} = (0, 1)$ .

## Notable exponential dispersion families

| Model                           | $N(\mu_i, \sigma^2)$ | $\text{Gamma}(\alpha, \alpha/\mu_i)$ | $\frac{1}{m_i} \text{Binomial}(m_i, \mu_i)$ | $\text{Poisson}(\mu_i)$ |
|---------------------------------|----------------------|--------------------------------------|---------------------------------------------|-------------------------|
| <b>Support</b> $\mathcal{Y}$    | $\mathbb{R}$         | $(0, \infty)$                        | $\{0, 1/m_i, \dots, 1\}$                    | $\{0, 1, 2, \dots\}$    |
| $\theta_i = \theta(\mu_i)$      | $\mu_i$              | $-1/\mu_i$                           | $\log\left(\frac{\mu_i}{1-\mu_i}\right)$    | $\log \mu_i$            |
| <b>Parameter space</b> $\Theta$ | $\mathbb{R}$         | $(-\infty, 0)$                       | $\mathbb{R}$                                | $\mathbb{R}$            |
| $b(\theta_i)$                   | $\theta_i^2/2$       | $-\log(-\theta_i)$                   | $\log\{1 + \exp(\theta_i)\}$                | $\exp(\theta_i)$        |
| $\phi$                          | $\sigma^2$           | $1/\alpha$                           | 1                                           | 1                       |
| $a_i(\phi)$                     | $\sigma^2$           | $1/\alpha$                           | $1/m_i$                                     | 1                       |
| $\mathcal{M}$                   | $\mathbb{R}$         | $(0, \infty)$                        | $(0, 1)$                                    | $(0, \infty)$           |
| $v(\mu_i)$                      | 1                    | $\mu_i^2$                            | $\mu_i(1 - \mu_i)$                          | $\mu_i$                 |

The list of exponential dispersion families does not end here. Other examples are the **inverse Gaussian**, the **negative binomial** and **hyperbolic secant** distributions.

# Link functions and canonical link

- To complete the GLM specification, we need to choose a **link function**  $g(\cdot)$  such that:

$$g(\mu_i) = \mathbf{x}_i^T \beta, \quad \theta_i = \theta(\mu_i) \implies \theta_i = \theta(g^{-1}(\mathbf{x}_i^T \beta)).$$

- It is fairly natural to consider a **monotone** and **differentiable** link function  $g(\cdot) : \mathcal{M} \rightarrow \mathbb{R}$  so that the inverse  $g^{-1}(\cdot) : \mathbb{R} \rightarrow \mathcal{M}$ . This ensures that the predictions are well-defined.

$$\mathbb{E}(Y_i) = g^{-1}(\mathbf{x}_i^T \beta) \in \mathcal{M}.$$

- For example, in **binary regression** any continuous **cumulative distribution function** for  $g^{-1}(\cdot)$  leads to a good link function, such as  $g^{-1}(\cdot) = \Phi(\cdot)$  (probit) or  $g^{-1}(\eta_i) = e^{\eta_i} / (1 + e^{\eta_i})$  (logistic).

The following link is called **canonical link** and it is implied by the distribution:

$$g(\mu_i) = \theta(\mu_i) \implies \theta_i = \mathbf{x}_i^T \beta.$$

- The **identity** link is canonical for the **Gaussian**, the **logarithm** is canonical for the **Poisson**, the **logit** is canonical for the Binomial and the **reciprocal** is canonical for the **Gamma**.

# Likelihood quantities

## Likelihood function

- Let  $Y_i \stackrel{\text{ind}}{\sim} \text{ED}(\mu_i, a_i(\phi)v(\mu_i))$  be the response variable of a GLM, with  $g(\mu_i) = \mathbf{x}_i^T \beta$ . The **joint distribution** of the responses  $\mathbf{Y} = (Y_1, \dots, Y_n)^T$  is

$$p(\mathbf{y}; \beta, \phi) = \prod_{i=1}^n p(y_i; \beta, \phi) = \prod_{i=1}^n \exp \left\{ \frac{y_i \theta_i - b(\theta_i)}{a_i(\phi)} + c(y_i, \phi) \right\}.$$

with  $\theta_i = \theta(\mu_i) = \theta(g^{-1}(\mathbf{x}_i^T \beta))$ .

- The **log-likelihood** function therefore is

$$\ell(\beta, \phi) = \sum_{i=1}^n \left\{ \frac{y_i \theta_i - b(\theta_i)}{a_i(\phi)} + c(y_i, \phi) \right\}.$$

- In general, there is **no sufficient statistic** with dimension smaller than  $n$ .

# Likelihood equations I

- To conduct inference using the **classical theory** (as in *Statistica II*), we need to consider the first and second derivative of the log-likelihood, that is, the **score function**

$$\ell_*(\beta, \phi) := \frac{\partial}{\partial \beta} \ell(\beta, \phi),$$

and the **observed information matrix**  $\mathbf{J}$ , whose elements are

$$j_{rs} = -\frac{\partial}{\partial \beta_r} \frac{\partial}{\partial \beta_s} \ell(\beta, \phi), \quad r, s = 1, \dots, p.$$

- These quantities have a **simple expression** in the end, but getting there requires quite a **bit of calculus**.

Let us focus on the estimation of  $\beta$ , assuming for now that  $\phi$  is a **known parameter**, as is the case in binomial or Poisson regression.

This assumption is not restrictive, even when  $\phi$  is actually unknown. In fact, we will show that the maximum likelihood estimate  $\hat{\beta}$  does not depend on  $\phi$ , and that  $\beta$  and  $\phi$  are **orthogonal**.

## Likelihood equations II

- Let us begin by noting that

$$\ell_r(\beta, \phi) = \frac{\partial}{\partial \beta_r} \ell(\beta, \phi) = \sum_{i=1}^n \frac{1}{a_i(\phi)} \left( y_i \frac{\partial \theta_i}{\partial \beta_r} - \frac{\partial b(\theta_i)}{\partial \beta_r} \right), \quad r = 1, \dots, p.$$

Such an expression can be **simplified** because

$$\frac{\partial b(\theta_i)}{\partial \beta_r} = b'(\theta_i) \frac{\partial \theta_i}{\partial \beta_r} = \mu_i \frac{\partial \theta_i}{\partial \beta_r},$$

which implies that the score function will have the following **structure**:

$$\frac{\partial}{\partial \beta_r} \ell(\beta, \phi) = \sum_{i=1}^n \frac{1}{a_i(\phi)} (y_i - \mu_i) \frac{\partial \theta_i}{\partial \beta_r}, \quad r = 1, \dots, p.$$

- Recall that  $a_i(\phi) = \phi/\omega_i$ , hence the **maximum likelihood estimator** is obtained by solving:

$$\cancel{\frac{1}{\phi}} \sum_{i=1}^n \omega_i (y_i - \mu_i) \frac{\partial \theta_i}{\partial \beta_r} = 0, \quad r = 1, \dots, p.$$

## Likelihood equations III

Let  $f(x)$  be a function with **inverse**  $g(x) = f^{-1}(x)$  and **first derivative**  $f'(x)$ . Then

$$\frac{\partial g}{\partial x} = [f^{-1}]'(x) = \frac{1}{f'(f^{-1}(x))}.$$

- Recall that  $g(\mu_i) = \mathbf{x}_i^T \boldsymbol{\beta} = \eta_i$  and that  $\theta_i = \theta(\mu_i)$  is the inverse of  $\mu(\theta_i)$ . As an **application** of the above **lemma**:

$$\frac{\partial \theta_i}{\partial \mu_i} = \theta'(\mu_i) = \frac{1}{\mu'(\theta(\mu_i))} = \frac{1}{b''(\theta(\mu_i))} = \frac{1}{v(\mu_i)},$$

Moreover, since  $\mu_i = g^{-1}(\eta_i)$  we obtain

$$\frac{\partial \mu_i}{\partial \eta_i} = \frac{1}{g'(g^{-1}(\eta_i))} = \frac{1}{g'(\mu_i)}.$$

- Summing up, the **chain rule of differentiation** for **composite functions** gives:

$$\frac{\partial \theta_i}{\partial \beta_r} = \frac{\partial \theta_i}{\partial \mu_i} \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta_r} = \frac{1}{v(\mu_i)} \frac{1}{g'(\mu_i)} x_{ir}, \quad r = 1, \dots, p.$$

## Likelihood equations IV

- Combining all the former equations, we obtain an explicit formula for the **score function**

$$\frac{\partial}{\partial \beta_r} \ell(\beta, \phi) = \frac{1}{\phi} \sum_{i=1}^n \omega_i \frac{(y_i - \mu_i)}{v(\mu_i)} \frac{x_{ir}}{g'(\mu_i)} = \sum_{i=1}^n \frac{(y_i - \mu_i)}{\text{var}(Y_i)} \frac{x_{ir}}{g'(\mu_i)}, \quad r = 1, \dots, p.$$

The **maximum likelihood estimator** solves the **likelihood equations**:

$$\sum_{i=1}^n \omega_i \frac{(y_i - \mu_i)}{v(\mu_i)} \frac{x_{ir}}{g'(\mu_i)} = 0, \quad r = 1, \dots, p,$$

which **do not depend** on  $\phi$ . In **matrix notation**

$$\mathbf{D}^T \mathbf{V}^{-1} (\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0},$$

where  $\mathbf{V} = \text{diag}(v(\mu_1)/\omega_1, \dots, v(\mu_n)/\omega_n)$  and  $\mathbf{D}$  is an  $n \times p$  matrix whose elements are

$$d_{ir} = \frac{\partial \mu_i}{\partial \beta_r} = \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta_r} = \frac{1}{g'(\mu_i)} x_{ir}, \quad i = 1, \dots, n, \quad r = 1, \dots, p.$$

## Canonical link: simplifications

- When using the **canonical link**  $\theta(\mu_i) = g(\mu_i)$  significant simplifications arise, because

$$\frac{\partial \theta_i}{\partial \mu_i} = \frac{1}{v(\mu_i)} = g'(\mu_i) \implies v(\mu_i)g'(\mu_i) = 1.$$

Thus, plugging this equality into the previous equations gives:

$$\frac{\partial \theta_i}{\partial \beta_r} = x_{ir}, \quad r = 1, \dots, p,$$

which is not surprising, because the canonical link implies  $\theta_i = x_{i1}\beta_1 + \dots + x_{ip}\beta_p$ .

The **likelihood equations** under the **canonical link** are

$$\sum_{i=1}^n \omega_i (y_i - \mu_i) x_{ir} = 0, \quad r = 1, \dots, p.$$

Let  $\mathbf{\Omega} = \text{diag}(\omega_1, \dots, \omega_n)$ , then in **matrix notation** we have  $\mathbf{X}^T \mathbf{\Omega} (\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}$ . The equations simplify even further when the weights are constant, i.e.  $\mathbf{\Omega} = I_n$ , yielding  $\mathbf{X}^T (\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}$ .

## Examples of estimating equations

Let  $Y_i \sim \text{ED}(\mu_i, \phi)$  with  $g(\mu_i) = \mu_i$ , namely the **Gaussian** linear model with the **identity** (canonical) link. The likelihood equations are

$$\mathbf{X}^T(\mathbf{y} - \mathbf{X}\beta) = \mathbf{0},$$

which are also called **normal equations**. Their solution over  $\beta$  is the OLS  $\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ .

Let  $Y_i \sim \text{ED}(\mu_i, \phi/\omega_i)$  with  $g(\mu_i) = \mu_i$ , namely the **Gaussian** linear model with the **identity** (canonical) link and **heteroscedastic errors**. The likelihood equations are

$$\mathbf{X}^T \mathbf{\Omega}(\mathbf{y} - \mathbf{X}\beta) = \mathbf{0},$$

Their solution over  $\beta$  is the weighted least squares estimator  $\hat{\beta}_{\text{wls}} = (\mathbf{X}^T \mathbf{\Omega} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{\Omega} \mathbf{y}$ .

Let  $Y_i \sim \text{ED}(\mu_i, \mu_i)$  with  $g(\mu_i) = \log \mu_i$ , namely a **Poisson** regression model with the **logarithmic** (canonical) link. The likelihood equations can be solved **numerically**

$$\mathbf{X}^T(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}, \quad \boldsymbol{\mu} = (e^{\mathbf{x}_1^T \beta}, \dots, e^{\mathbf{x}_n^T \beta})^T.$$

## Example: Beetles data

- Using the **Beetles** data, we specified a **binomial logistic** regression model for the counts  $m_i Y_i \sim \text{Binomial}(m_i, \pi_i)$  with mean  $\mathbb{E}(Y_i) = \pi_i = \exp(\beta_1 + \beta_2 x_i) / (1 + \exp(\beta_1 + \beta_2 x_i))$ .
- The **maximum likelihood estimate**  $(\hat{\beta}_1, \hat{\beta}_2)$  is the value solving simultaneously:

$$\sum_{i=1}^n m_i y_i = \sum_{i=1}^n m_i \frac{\exp(\beta_1 + \beta_2 x_i)}{1 + \exp(\beta_1 + \beta_2 x_i)}, \quad \text{and} \quad \sum_{i=1}^n m_i x_i y_i = \sum_{i=1}^n m_i x_i \frac{\exp(\beta_1 + \beta_2 x_i)}{1 + \exp(\beta_1 + \beta_2 x_i)}.$$

Unfortunately, there is **no closed form** solution.

- In our case, we have that

$$\sum_{i=1}^n m_i y_i = 291, \quad \sum_{i=1}^n m_i x_i y_i = 532.2083.$$

- With these values, we can use the **numerical** algorithm IRLS to solve the above system, obtaining

$$\hat{\beta} = (\hat{\beta}_1, \hat{\beta}_2) = (-60.717, 34.270).$$

## Example: Beetles data

- The **predicted response** can be computed by using the formula

$$\hat{\mu}_i = \frac{\exp(\hat{\beta}_1 + \hat{\beta}_2 x_i)}{1 + \exp(\hat{\beta}_1 + \hat{\beta}_2 x_i)} = \frac{\exp(-60.717 + 34.270 x_i)}{1 + \exp(-60.717 + 34.270 x_i)}, \quad i = 1, \dots, 8.$$

| $m_i$ | <b>deaths</b> ( $S_i$ ) | <b>logdose</b> ( $x_i$ ) | $Y_i = S_i/m_i$ | $\hat{\mu}_i$ |
|-------|-------------------------|--------------------------|-----------------|---------------|
| 59    | 6                       | 1.691                    | 0.102           | 0.059         |
| 60    | 13                      | 1.724                    | 0.217           | 0.164         |
| 62    | 18                      | 1.755                    | 0.290           | 0.362         |
| 56    | 28                      | 1.784                    | 0.500           | 0.605         |
| 63    | 52                      | 1.811                    | 0.825           | 0.795         |
| 59    | 53                      | 1.837                    | 0.898           | 0.903         |
| 62    | 61                      | 1.861                    | 0.984           | 0.955         |
| 60    | 60                      | 1.884                    | 1.000           | 0.979         |

- The predicted values and the data  $Y_i$  were also shown in a **plot** at the **beginning of this unit**.

## Example: Aids data

- In the **Aids** data, we specified a **Poisson** regression model with  $\mathbb{E}(Y_i) = \exp(\beta_1 + \beta_2 x_i)$ .
- The maximum likelihood estimate  $(\hat{\beta}_1, \hat{\beta}_2)$  **solves** simultaneously:

$$\sum_{i=1}^n y_i = \sum_{i=1}^n \exp(\beta_1 + \beta_2 x_i), \quad \text{and} \quad \sum_{i=1}^n x_i y_i = \sum_{i=1}^n x_i \exp(\beta_1 + \beta_2 x_i).$$

- This system does **not** always admit a **solution**. This happens, for example, in the **extreme case**  $\sum_{i=1}^n y_i = 0$ , occurring when all counts equal zero.
- Using the **Aids** data we have  $\sum_{i=1}^n y_i = 217$  and  $\sum_{i=1}^n x_i y_i = 2387$ . Via **numerical methods** we solve the above system of equations and we obtain  $\hat{\beta}_1 = 0.304$  and  $\hat{\beta}_2 = 0.259$ .
- The estimated mean values are  $\hat{\mu}_i = \exp(0.304 + 0.259x_i)$  and in particular the mean for the **next period** is

$$\hat{\mu}_{i+1} = \exp(0.304 + 0.259(x_i + 1)) = \exp(0.259)\hat{\mu}_i = 1.296\hat{\mu}_i.$$

In other words, the estimated number of deaths increases by about 30% every quarter.

## Example: Aids data

|        | deaths ( $Y_i$ ) | period ( $x_i$ ) | $\hat{\mu}_i$ |
|--------|------------------|------------------|---------------|
| 1983-1 | 0                | 1                | 1.755         |
| 1983-2 | 1                | 2                | 2.274         |
| 1983-3 | 2                | 3                | 2.946         |
| 1983-4 | 3                | 4                | 3.817         |
| 1984-1 | 1                | 5                | 4.945         |
| 1984-2 | 4                | 6                | 6.407         |
| 1984-3 | 8                | 7                | 8.301         |
| 1984-4 | 17               | 8                | 10.755        |
| 1985-1 | 23               | 9                | 13.934        |
| 1985-2 | 32               | 10               | 18.052        |
| 1985-3 | 20               | 11               | 23.389        |
| 1985-4 | 24               | 12               | 30.302        |
| 1986-1 | 37               | 13               | 39.259        |
| 1986-2 | 45               | 14               | 50.863        |

- The predicted values and the data  $Y_i$  were also shown in a **plot** at the **beginning of this unit**.

## Observed and expected information I

- Let us first consider the negative derivative of the score function, that is the **observed information matrix**  $\mathbf{J}$  with entries:

$$\begin{aligned} j_{rs} &= -\frac{\partial}{\partial \beta_s} \left[ \frac{\partial}{\partial \beta_r} \ell(\beta, \phi) \right] = -\frac{\partial}{\partial \beta_s} \sum_{i=1}^n \frac{1}{a_i(\phi)} (y_i - \mu_i) \frac{\partial \theta_i}{\partial \beta_r} \\ &= \sum_{i=1}^n \frac{1}{a_i(\phi)} \left[ \frac{\partial \mu_i}{\partial \beta_s} \frac{\partial \theta_i}{\partial \beta_r} - (y_i - \mu_i) \frac{\partial^2 \theta_i}{\partial \beta_r \partial \beta_s} \right], \quad r, s = 1, \dots, p. \end{aligned}$$

- Let  $\mathbf{I} = \mathbb{E}(\mathbf{J})$  be the  $p \times p$  **Fisher information matrix** associated with  $\beta$ , whose elements are

$$i_{rs} = \mathbb{E}(j_{rs}) = \mathbb{E} \left( -\frac{\partial}{\partial \beta_r} \frac{\partial}{\partial \beta_s} \ell(\beta, \phi) \right), \quad r, s = 1, \dots, p.$$

- Thus, the Fisher information matrix substantially simplifies because  $\mathbb{E}(Y_i) = \mu_i$ , obtaining:

$$i_{rs} = \sum_{i=1}^n \frac{1}{a_i(\phi)} \frac{\partial \mu_i}{\partial \beta_s} \frac{\partial \theta_i}{\partial \beta_r}, \quad r, s = 1, \dots, p.$$

## Observed and expected information II

- In the previous slides we already computed the explicit values of these derivatives:

$$\frac{\partial \mu_i}{\partial \beta_s} = \frac{x_{is}}{g'(\mu_i)}, \quad \frac{\partial \theta_i}{\partial \beta_r} = \frac{x_{ir}}{v(\mu_i)g'(\mu_i)}.$$

Combining the above equations, we obtain that the **Fisher information**  $\mathbf{I}$  of a GLM has entries

$$i_{rs} = \frac{1}{\phi} \sum_{i=1}^n \omega_i \frac{x_{ir}x_{is}}{v(\mu_i)g'(\mu_i)^2} = \sum_{i=1}^n \frac{x_{ir}x_{is}}{\text{var}(Y_i)g'(\mu_i)^2}, \quad r, s = 1, \dots, p.$$

In **matrix notation**, we have that

$$\mathbf{I} = \mathbf{X}^T \mathbf{W} \mathbf{X},$$

where  $\mathbf{W} = \text{diag}(w_1, \dots, w_n)$  and  $w_i$  are **weights** such that

$$w_i = \frac{1}{\phi} \frac{\omega_i}{v(\mu_i)g'(\mu_i)^2} = \frac{1}{\text{var}(Y_i)g'(\mu_i)^2}, \quad i = 1, \dots, n.$$

## Canonical link: simplifications

Under the **canonical link** we have that  $\theta_i = x_{i1}\beta_1 + \dots + \beta_px_{ip}$ , which means that

$$\frac{\partial^2 \theta_i}{\partial \beta_r \partial \beta_s} = 0 \quad \implies \quad i_{rs} = j_{rs} = \sum_{i=1}^n \frac{1}{a_i(\phi)} \frac{\partial \mu_i}{\partial \beta_s} \frac{\partial \theta_i}{\partial \beta_r}.$$

The **observed information**  $\mathbf{J}$  is **non-stochastic**, which means that observed information and expected (Fisher) information coincide, that is  $i_{rs} = j_{rs}$  and  $\mathbf{I} = \mathbf{J}$ .

Under the **canonical link**, we also have the simplifications  $1/v(\mu_i) = g'(\mu_i)$ , yielding

$$i_{rs} = \frac{1}{\phi} \sum_{i=1}^n \omega_i v(\mu_i) x_{ir} x_{is}, \quad r, s = 1, \dots, p.$$

In **matrix notation**, we have that  $\mathbf{I} = \mathbf{X}^T \mathbf{W} \mathbf{X}$  with weights

$$w_i = \frac{1}{\phi} \omega_i v(\mu_i) = \frac{v(\mu_i)}{a_i(\phi)}, \quad i = 1, \dots, n.$$

## Further considerations

- The **observed** and **expected** information matrices  $\mathbf{J}$  and  $\mathbf{I}$ , as well as weights  $\mathbf{W}$ , **depend** on  $\beta$  and  $\phi$ . We write  $\hat{\mathbf{J}}$ ,  $\hat{\mathbf{I}}$  and  $\hat{\mathbf{W}}$  to indicate that  $\beta$  and  $\phi$  have been estimated with  $\hat{\beta}$  and  $\hat{\phi}$ .
- If  $\mathbf{X}$  has **full rank** and  $g'(\mu) \neq 0$ , then  $\mathbf{I}$  is **positive definite** for any value of  $\beta$  and  $\phi$ .
- Under the **canonical link**, we have  $\mathbf{J} = \mathbf{I}$ , and both matrices are **positive definite** if  $\text{rk}(\mathbf{X}) = p$ .
- This implies that the log-likelihood function is **concave** because its second derivative is negative definite, so any **solution** to the **estimating equations** is also a **global optimum**.
- The Fisher information matrix could be computed exploiting **Bartlett identity**, namely

$$i_{rs} = \mathbb{E} \left[ \left( \frac{\partial}{\partial \beta_r} \ell(\beta, \phi) \right) \left( \frac{\partial}{\partial \beta_s} \ell(\beta, \phi) \right) \right], \quad r, s = 1, \dots, p.$$

as in Agresti (2015). Of course, the final result **coincides** with ours.

## - Orthogonality of $\beta$ and $\phi$

- Let us now consider the case in which  $\phi$  is **unknown** so that  $a_i(\phi) = \phi/\omega_i$ . We obtain:

$$j_{r\phi} = -\frac{\partial}{\partial \beta_r} \frac{\partial}{\partial \phi} \ell(\beta, \phi) = \frac{1}{\phi^2} \sum_{i=1}^n \omega_i (y_i - \mu_i) \frac{\partial \theta_i}{\partial \beta_r}, \quad r = 1, \dots, p.$$

whose **expected value** is  $i_{r\phi} = \mathbb{E}(j_{r\phi}) = 0$  since  $\mathbb{E}(Y_i) = \mu_i$ .

- This means the Fisher information matrix accounting for  $\phi$  takes the form:

$$\begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & i_{\phi\phi} \end{pmatrix} \implies \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & i_{\phi\phi} \end{pmatrix}^{-1} = \begin{pmatrix} \mathbf{I}^{-1} & \mathbf{0} \\ \mathbf{0} & 1/i_{\phi\phi} \end{pmatrix}$$

where  $[\mathbf{I}]_{rs} = i_{rs}$  are the elements associated to  $\beta$  as before.

The parameters  $\beta$  and  $\phi$  are **orthogonal** and their estimates are **asymptotically independent**.

Moreover, the matrices  $\mathbf{I}$  and  $\mathbf{I}^{-1}$  are all that is needed for inference on  $\beta$ , so there is no need to compute  $i_{\phi\phi}$ . Moreover, the maximum likelihood  $\hat{\beta}$  can also be computed without knowing  $\phi$ .

# IRLS algorithm

# Numerical methods for maximum likelihood estimation

- In general, the estimating equations of a GLM

$$\mathbf{D}^T \mathbf{V}^{-1}(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}$$

cannot be solved in **closed form** and we need to rely on **numerical methods**.

- An **iterative** method means that we start the **algorithm** with a candidate value  $\beta^{(1)}$  (**initialization**). Then, at the step  $t$  we update

$$\beta^{(t+1)} = \text{update}(\beta^{(t)}), \quad t = 1, 2, \dots$$

- The algorithm **stops** whenever a certain criterion is met, e.g. when  $\|\beta^{(t+1)} - \beta^{(t)}\| < \epsilon$ , where  $\epsilon$  is sometimes called **tolerance**. We say it reached **convergence**.
- The **iteratively re-weighted least squares** (IRLS) algorithm became very popular after being proposed by Nelder and Wedderburn (1972) and is currently implemented in **R**.
- The IRLS algorithm can be used for any GLM, has a clear geometric interpretation, and often delivers good performance. It can be seen as a **variant** of **Newton-Raphson**.

# Newton-Raphson algorithm I

- In the Newton-Raphson algorithm, we consider a second-order **Taylor expansion** of the log-likelihood  $\ell(\beta) = \ell(\beta, \phi)$  centered in  $\beta^{(t)}$ , namely:

$$\ell(\beta) \approx \ell(\beta^{(t)}) + \ell_*(\beta^{(t)})^T (\beta - \beta^{(t)}) - \frac{1}{2} (\beta - \beta^{(t)})^T \mathbf{J}^{(t)} (\beta - \beta^{(t)})$$

where  $\ell_*(\beta^{(t)})$  is the **score function** and  $\mathbf{J}^{(t)}$  is the **observed information**, evaluated at  $\beta^{(t)}$ .

- In other words, we **approximate** the log-likelihood  $\ell(\beta)$  with a **parabola**. This gives the **approximate likelihood equations**:

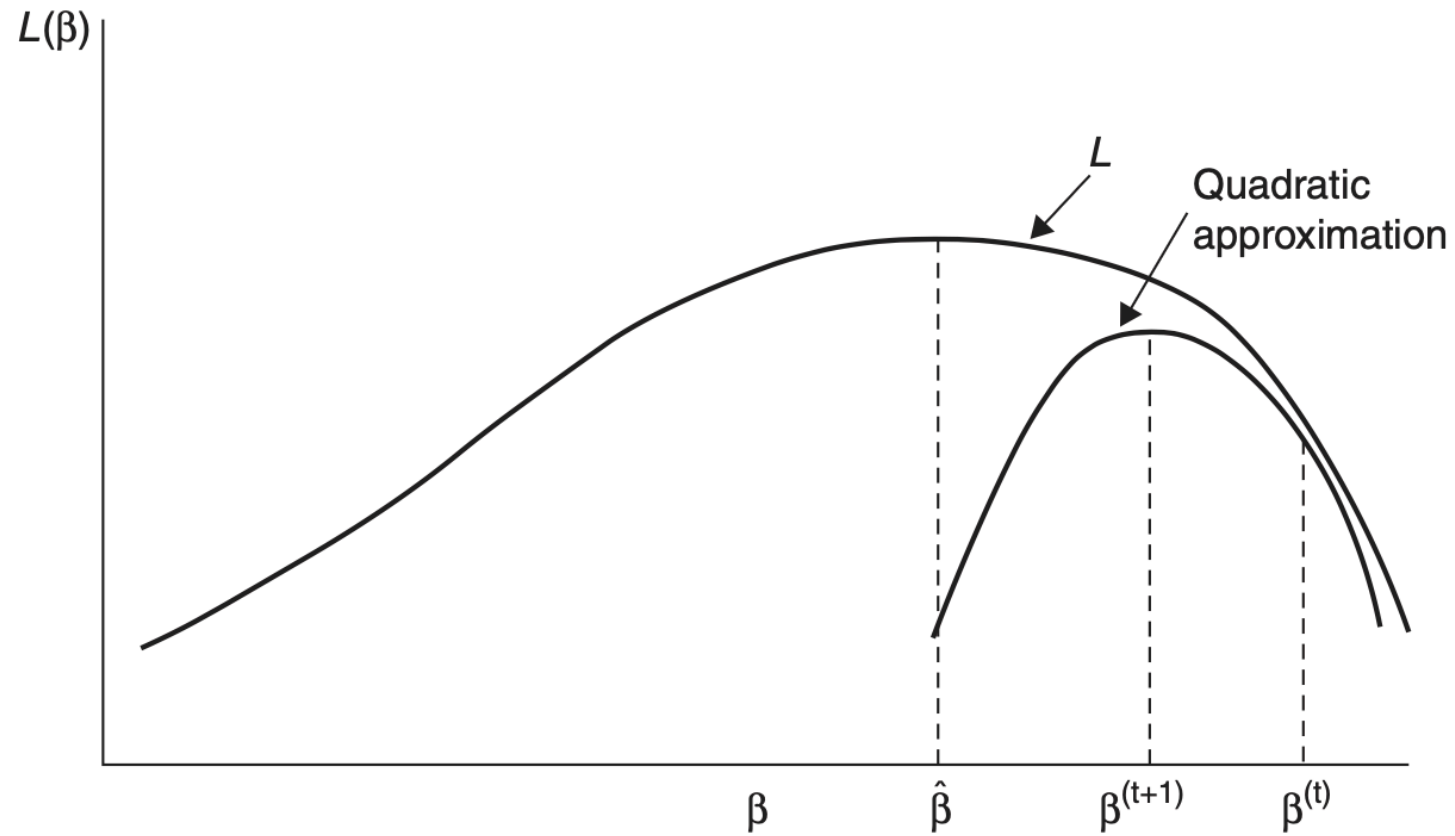
$$\ell_*(\beta^{(t)}) - \mathbf{J}^{(t)} (\beta - \beta^{(t)}) = \mathbf{0}.$$

- Solving the equation above gives the following **updates**:

$$\beta^{(t+1)} = \beta^{(t)} + (\mathbf{J}^{(t)})^{-1} \ell_*(\beta^{(t)}), \quad t = 1, 2, \dots$$

The Newton-Raphson method essentially considers a series of **parabolic approximations** to the log-likelihood, each time evaluating the **point of maximum**.

## Newton-Raphson algorithm II



**Figure 4.2** Illustration of a cycle of the Newton–Raphson method.

- Figure taken from Agresti (2015).

# Iteratively re-weighted least squares I

- The matrix  $\mathbf{J}^{(t)}$  is not always invertible, therefore the algorithm may crash. To remedy this, we replace it with the **expected information**  $\mathbf{I}^{(t)}$ .

In the **iteratively re-weighted least squares** (IRLS) algorithm, we consider the updates:

$$\beta^{(t+1)} = \beta^{(t)} + (\mathbf{I}^{(t)})^{-1} \ell_*(\beta^{(t)}), \quad t = 1, 2, \dots$$

This method is also called **Fisher scoring**.

- The above formula can be **simplified** a bit. First, we rewrite the **score** as

$$\frac{\partial}{\partial \beta_r} \ell(\beta, \phi) = \frac{1}{\phi} \sum_{i=1}^n \omega_i \frac{(y_i - \mu_i)}{v(\mu_i)} \frac{x_{ir}}{g'(\mu_i)} = \sum_{i=1}^n x_{ir} w_i (y_i - \mu_i) g'(\mu_i),$$

where the **weights** were defined as  $w_i = \omega_i / (\phi v(\mu_i) g'(\mu_i)^2)$ . In **matrix notation** we will write:

$$\ell_*(\beta^{(t)}) = \mathbf{X}^T \mathbf{W}^{(t)} \mathbf{u}^{(t)}, \quad \mathbf{I}^{(t)} = \mathbf{X}^T \mathbf{W}^{(t)} \mathbf{X},$$

where  $\mathbf{u}^{(t)} = (u_1^{(t)}, \dots, u_n^{(t)})^T$  and  $u_i^{(t)} = (y_i - \mu_i^{(t)}) g'(\mu_i^{(t)})$  for  $i = 1, \dots, n$ .

## Iteratively re-weighted least squares II

- Exploiting the above formulas, we can write the IRLS update as follows

$$\beta^{(t+1)} = \beta^{(t)} + (\mathbf{X}^T \mathbf{W}^{(t)} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W}^{(t)} \mathbf{u}^{(t)}.$$

Now **multiply** both sides by  $(\mathbf{X}^T \mathbf{W}^{(t)} \mathbf{X})$ , **simplify** and **re-arrange** the resulting terms. This gives the following formula.

In the **iteratively re-weighted least squares** (IRLS) algorithm, we consider the updates:

$$\beta^{(t+1)} = (\mathbf{X}^T \mathbf{W}^{(t)} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W}^{(t)} \mathbf{z}^{(t)}, \quad t = 1, 2, \dots,$$

where  $\mathbf{z}^{(t)} = (z_1^{(t)}, \dots, z_n^{(t)})^T$  is called **pseudo-response** whose elements are defined as

$$z_i^{(t)} = \mathbf{x}_i^T \beta^{(t)} + u_i^{(t)} = \mathbf{x}_i^T \beta^{(t)} + (y_i - \mu_i^{(t)}) g'(\mu_i^{(t)}), \quad i = 1, \dots, n.$$

Hence, each update can be interpreted as the solution of a **weighted least squares** problem:

$$\beta^{(t+1)} = \arg \min_{\beta \in \mathbb{R}^p} (\mathbf{z}^{(t)} - \mathbf{X} \beta)^T \mathbf{W}^{(t)} (\mathbf{z}^{(t)} - \mathbf{X} \beta).$$

## Iteratively re-weighted least squares III

- The IRLS update does not depend on the choice of  $\phi$ , because it **cancels** in the multiplications, as we would expect.
- The pseudo-responses have a **nice interpretation**, because they can be interpreted as a linear approximation of the transformed responses:

$$g(y_i) \approx g(\mu_i) + (y_i - \mu_i)g'(\mu_i) = \eta_i + (y_i - \mu_i)g'(\mu_i) = z_i.$$

- Based on this approximation, a good **initialization** is obtained by setting  $\mathbf{W} = \mathbf{I}_n$  and  $z_i = g(y_i)$ , which gives

$$\beta^{(1)} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T g(\mathbf{y}),$$

namely the **least squares** solution for the transformed data. To avoid boundary issues, sometimes the data are perturbed, as we did in binomial regression.

## Example: IRLS for logistic regression

- Consider a **logistic regression** model for **proportions**  $Y_i \in \{0, 1/m_i, \dots, 1\}$  with probability of success  $\pi_i = \mu_i$  and trials  $m_i$ .

### IRLS algorithm for logistic regression

- Initialize**  $\beta^{(1)} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \text{logit}(\tilde{\mathbf{y}})$  where  $\text{logit}(\tilde{\mathbf{y}})$  is the **empirical logit** transform.
- For  $t = 1, 2, \dots$  until **convergence** compute:

$$\beta^{(t+1)} = (\mathbf{X}^T \mathbf{W}^{(t)} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W}^{(t)} \mathbf{z}^{(t)},$$

where the weights in  $\mathbf{W}^{(t)}$  equals  $w_i^{(t)} = m_i \pi_i^{(t)} (1 - \pi_i^{(t)})$  and the **pseudo-responses**  $\mathbf{z}^{(t)}$  are

$$z_i^{(t)} = \mathbf{x}_i^T \beta^{(t)} + \frac{y_i - \pi_i^{(t)}}{\pi_i^{(t)} (1 - \pi_i^{(t)})}, \quad i = 1, \dots, n,$$

with probabilities  $\pi_i^{(t)} = \exp(\mathbf{x}_i^T \beta^{(t)}) / (1 + \exp(\mathbf{x}_i^T \beta^{(t)}))$  for  $i = 1, \dots, n$ .

## Estimation of the dispersion $\phi$

- In some GLMs, such as the Gaussian and the Gamma, there is a **dispersion parameter**  $\phi$  that we need to estimate.
- Because of **numerical instabilities** and **lack of robustness**, a **method of moments** estimator is typically preferred to the maximum likelihood one. If  $\mu_i$  were known, the estimator

$$\frac{1}{n} \sum_{i=1}^n \omega_i \frac{(y_i - \mu_i)^2}{v(\mu_i)}$$

would be **unbiased** for  $\phi$ , because  $\mathbb{E}\{(Y_i - \mu_i)^2\} = (\phi/\omega_i)v(\mu_i)$ . This motivates the estimator

$$\hat{\phi} = \frac{1}{n-p} \sum_{i=1}^n \omega_i \frac{(y_i - \hat{\mu}_i)^2}{v(\hat{\mu}_i)}, \quad \hat{\mu}_i = g^{-1}(\mathbf{x}_i^T \hat{\beta}).$$

- This is a **consistent** estimator of  $\phi$  as long as  $\hat{\beta}$  is consistent.
- When  $g(\mu_i) = \mu_i$  is the **identity link** and  $v(\mu_i) = \omega_i = 1$ , this coincides with the usual unbiased estimator  $s^2$  of  $\sigma^2$  for a Gaussian linear model.

# Inference and hypothesis testing

# Asymptotic distribution of $\hat{\beta}$

The **asymptotic distribution** of the **maximum likelihood estimator** is

$$\hat{\beta} \sim N_p \left( \beta, (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \right),$$

for large values of  $n$  and under mild regularity conditions on  $\mathbf{X}$ .

- Under correct specification and mild conditions on  $\mathbf{X}$ , the maximum likelihood estimator is **asymptotically unbiased** and with known asymptotic **variance**

$$\mathbb{E}(\hat{\beta} - \beta) \approx 0, \quad \text{var}(\hat{\beta}) \approx (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1}.$$

- In practice, since  $\mathbf{W}$  depends on  $\beta$  and  $\phi$ , we rely on the following approximation

$$\widehat{\text{var}}(\hat{\beta}) = (\mathbf{X}^T \hat{\mathbf{W}} \mathbf{X})^{-1},$$

where we plugged in the estimates  $\hat{\beta}$  and  $\hat{\phi}$  into  $\mathbf{W}$  obtaining  $\hat{\mathbf{W}}$ . The **standard errors** are:

$$\text{Std. Error} = [\widehat{\text{se}}(\hat{\beta})]_j = \sqrt{[(\mathbf{X}^T \hat{\mathbf{W}} \mathbf{X})^{-1}]_{jj}}$$

## Example: Beetles data

- Using the **Beetles** data, we specified a **binomial logistic** regression model for the counts  $m_i Y_i \sim \text{Binomial}(m_i, \pi_i)$  with mean  $\mu_i = \exp(\beta_1 + \beta_2 x_i) / (1 + \exp(\beta_1 + \beta_2 x_i))$ .
- We previously **estimated**  $\hat{\beta} = (-60.717, 34.270)$ . This means that the **weights** are estimated as

$$\hat{\mathbf{W}} = \text{diag}(m_1 \hat{\mu}_1 (1 - \hat{\mu}_1), \dots, m_n \hat{\mu}_n (1 - \hat{\mu}_n)) = \text{diag}(3.255, 8.227, \dots, 1.231).$$

from which we obtain the **estimated Fisher information matrix**:

$$\mathbf{X}^T \hat{\mathbf{W}} \mathbf{X} = \begin{pmatrix} \sum_{i=1}^n m_i \hat{\mu}_i (1 - \hat{\mu}_i) & \sum_{i=1}^n x_i m_i \hat{\mu}_i (1 - \hat{\mu}_i) \\ \sum_{i=1}^n x_i m_i \hat{\mu}_i (1 - \hat{\mu}_i) & \sum_{i=1}^n x_i^2 m_i \hat{\mu}_i (1 - \hat{\mu}_i) \end{pmatrix} = \begin{pmatrix} 58.484 & 104.011 \\ 104.011 & 185.095 \end{pmatrix}.$$

- Hence, the **estimated covariance** matrix of the maximum likelihood estimator is

$$\widehat{\text{var}}(\hat{\beta}) = (\mathbf{X}^T \hat{\mathbf{W}} \mathbf{X})^{-1} = \begin{pmatrix} 26.840 & -15.082 \\ -15.082 & 8.481 \end{pmatrix}.$$

- Therefore the **estimated standard errors** are

$$[\widehat{\text{se}}(\hat{\beta})]_j = \sqrt{[(\mathbf{X}^T \hat{\mathbf{W}} \mathbf{X})^{-1}]_{jj}} \implies \widehat{\text{se}}(\hat{\beta}) = (5.181, 2.912).$$

## Example: Aids data

- In the **Aids** data, we specified a **Poisson** regression model with  $\mathbb{E}(Y_i) = \exp(\beta_1 + \beta_2 x_i)$  and estimated  $\hat{\beta} = (0.304, 0.259)$ .
- This means that the weights are estimated as

$$\hat{\mathbf{W}} = \text{diag}(\hat{\mu}_1, \dots, \hat{\mu}_n) = \text{diag}(1.755, \dots, 50.863).$$

from which we obtain the **estimated Fisher information matrix**:

$$\mathbf{X}^T \hat{\mathbf{W}} \mathbf{X} = \begin{pmatrix} \sum_{i=1}^n \hat{\mu}_i & \sum_{i=1}^n x_i \hat{\mu}_i \\ \sum_{i=1}^n x_i \hat{\mu}_i & \sum_{i=1}^n x_i^2 \hat{\mu}_i \end{pmatrix} = \begin{pmatrix} 217 & 2387 \\ 2387 & 28279.05 \end{pmatrix}.$$

- Hence, the **estimated covariance** matrix of the maximum likelihood estimator is

$$\widehat{\text{var}}(\hat{\beta}) = (\mathbf{X}^T \hat{\mathbf{W}} \mathbf{X})^{-1} = \begin{pmatrix} 0.06445 & -0.00544 \\ -0.00544 & 0.00049 \end{pmatrix}.$$

- Therefore the **estimated standard errors** are

$$[\widehat{\text{se}}(\hat{\beta})]_j = \sqrt{[(\mathbf{X}^T \hat{\mathbf{W}} \mathbf{X})^{-1}]_{jj}} \implies \widehat{\text{se}}(\hat{\beta}) = (0.254, 0.022).$$

## Wald test and confidence intervals

- Consider the **hypothesis**  $H_0 : \beta_j = \beta_0$  against the **alternative**  $H_1 : \beta_j \neq \beta_0$ . The **Wald test statistic**  $z_j$ , rejecting the hypothesis for large values of  $|z_j|$  is:

$$\mathbf{z\ value} = z_j = \frac{\hat{\beta}_j - \beta_0}{[\widehat{\text{se}}(\hat{\beta})]_j} = \frac{\hat{\beta}_j - \beta_0}{\sqrt{[(\mathbf{X}^T \hat{\mathbf{W}} \mathbf{X})^{-1}]_{jj}}} \sim \text{N}(0, 1).$$

which is **approximately** distributed as a **standard normal** under  $H_0$ .

- The **p-value** is defined in the usual way, namely

$$\alpha_{\text{obs}} = \mathbb{P}(|Z| \geq |z_j|) = 2(1 - \Phi(|z_j|)), \quad Z \sim \text{N}(0, 1).$$

- By inverting the Wald test, we obtain the associated **confidence interval**

$$\hat{\beta}_j \pm z_{1-\alpha/2} \sqrt{[(\mathbf{X}^T \hat{\mathbf{W}} \mathbf{X})^{-1}]_{jj}}.$$

of **approximate** level  $1 - \alpha$ , where  $z_{1-\alpha/2}$  is the quantile of a standard Gaussian.

## Comparison with the Gaussian linear model

- In a classical **Gaussian linear model** the weight matrix is  $\mathbf{W} = I_n/\sigma^2$ , therefore

$$\hat{\beta} \sim N_p \left( \beta, \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1} \right).$$

- The **Wald statistic**  $z_j$  **specializes** to

$$z_j = \frac{\hat{\beta}_j - \beta_0}{[\widehat{\text{se}}(\hat{\beta})]_j} = \frac{\hat{\beta}_j - \beta_0}{s \sqrt{[(\mathbf{X}^T \mathbf{X})^{-1}]_{jj}}},$$

which is the usual test statistic considered, e.g., in the output of **lm** in **R**.

- However, in the Gaussian case there is **no need** of **approximations**. The distribution of  $z_j$  is a **Student's**  $t_{n-p}$  under  $H_0$ , which indeed converges to a  $N(0, 1)$  for large values of  $n$ .

In GLMs we use procedures that are **approximate** rather than **exact**. Of course, whenever an exact result is known, we should use it.

## Example: Beetles data

- The Wald test is the **default** choice in **R** for checking the hypotheses  $H_0 : \beta_j = 0$ . In the **Beetles** data we get the following familiar summary:

```
z test of coefficients:

              Estimate Std. Error z value Pr(>|z|)
(Intercept) -60.7175      5.1807 -11.720 < 2.2e-16 ***
logdose      34.2703      2.9121  11.768 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- Many of the above quantities (estimates and standard errors) have been obtained before.
- In this case, we **reject** the null hypothesis that  $\beta_2 = 0$ . Indeed, even from the scatterplot there was evidence of a relationship between the **deaths** proportion and the **logdose**.
- For completeness, we also compute the associated **Wald confidence intervals**, which are:

```
              2.5 %      97.5 %
(Intercept) -70.87144 -50.56347
logdose      28.56265  39.97800
```

## Example: Aids data

- The Wald tests for checking the hypotheses  $H_0 : \beta_j = 0$  in the Aids data are provided below.

z test of coefficients:

```

              Estimate Std. Error z value Pr(>|z|)
(Intercept) 0.303655   0.253867  1.1961   0.2317
period      0.258963   0.022238 11.6448   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

- In this case, we **reject** the null hypothesis that  $\beta_2 = 0$  because the **p-value**  $\Pr(>|z|) \approx 0$ . Again, this is not very surprising: the number of **deaths** was clearly increasing over time.
- The associated **Wald confidence intervals** are:

```

              2.5 %    97.5 %
(Intercept) -0.1939158 0.8012249
period      0.2153764 0.3025494

```

# General hypothesis testing

- Suppose we wish to test **multiple parameters** at the same time. Let us organize the parameters into **two blocks**:

$$\beta = \begin{pmatrix} \beta_A \\ \beta_B \end{pmatrix}, \quad \beta_A = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_{p_0} \end{pmatrix}, \quad \beta_B = \begin{pmatrix} \beta_{p_0+1} \\ \vdots \\ \beta_p \end{pmatrix},$$

where  $q = p - p_0$  is the number of **constrained parameters**. We want to **test the hypothesis**:

$$H_0 : \beta_B = \beta_0 \quad \text{against} \quad H_1 : \beta_B \neq \beta_0.$$

A common case is  $H_0 : \beta_B = 0$  (**nested models**), where we compare the **reduced model**  $M_0$  against the **full model**  $M_1$ . We verify if **all** the  $q$  variables associated with  $\beta_B$  can be omitted.

The case  $q = 1$ , that is  $\beta_B = \beta_p$  with  $H_0 : \beta_p = 0$  corresponds to the previously considered situation where we test if a **specific coefficient**, say  $\beta_p$ , is non-zero.

# Testing hypothesis in GLMs I

- There are **three classical tests** that we could consider for such a testing problem: the **Wald test**  $W_e$ , the **Rao-score test**  $W_u$ , and the **log-likelihood ratio test**  $W$ .
- All these tests **reject** the null hypothesis for **large values** of the **statistic**.

## Wald test (general case)

Let  $\hat{\beta} = (\hat{\beta}_A, \hat{\beta}_B)$  be the **unrestricted** maximum likelihood estimate; the quantity

$$W_e = (\hat{\beta}_B - \beta_0)^T \widehat{\text{var}}(\hat{\beta}_B)^{-1} (\hat{\beta}_B - \beta_0),$$

is called **Wald test**. Here  $\widehat{\text{var}}(\hat{\beta}_B)$  is the appropriate **block** of  $(\mathbf{X}^T \hat{\mathbf{W}} \mathbf{X})^{-1}$  and  $\hat{\mathbf{W}}$  is estimated using  $\hat{\beta}$  and  $\hat{\phi}$ . Under  $H_0$  this quantity is **approximately** distributed as

$$W_e \sim \chi_q^2,$$

a  $\chi^2$  distribution with  $q$  **degrees of freedom**. The **p-value** is  $\Pr(>\text{Chi}) = \mathbb{P}(W_e > w_{e, \text{obs}})$ .

- Clearly, in the  $q = 1$  case we recover the Wald statistic with  $z_j^2 = W_e$ .

# Log-likelihood ratio test

## Log-likelihood ratio test (LRT)

Let  $\hat{\beta} = (\hat{\beta}_A, \hat{\beta}_B)$  be the **unrestricted** maximum likelihood estimate and let  $\hat{\beta}_0 = (\hat{\beta}_{A,0}, \hat{\beta}_0)$  be the **restricted** maximum likelihood estimate. The quantity

$$W = 2[\ell(\hat{\beta}, \hat{\phi}) - \ell(\hat{\beta}_0, \hat{\phi})],$$

is called **log-likelihood ratio** test (LRT). Under  $H_0$  this quantity is **approximately** distributed as

$$W \sim \chi_q^2,$$

a  $\chi^2$  distribution with  $q$  **degrees of freedom**. The **p-value** is  $\Pr(>\text{Chi}) = \mathbb{P}(W > w_{\text{obs}})$ .

- When testing  $H_0 : \beta_B = 0$ , we separately fit the **full model**, obtaining  $\hat{\beta}$ , and the **reduced model**, obtaining  $\hat{\beta}_0 = (\hat{\beta}_{A,0}, 0)$ . Then, we compare their log-likelihoods:  $\ell(\hat{\beta}, \hat{\phi}) - \ell(\hat{\beta}_0, \hat{\phi})$ .
- The LRT is the **default** in **R** for comparing **nested models**.

When the dispersion parameter  $\phi$  is unknown, a **variant** uses **separate estimates**  $\hat{\phi}$ , based on  $\hat{\beta}$ , and  $\hat{\phi}_0$ , based on  $\hat{\beta}_0$ . The **anova** **R** command uses a single  $\hat{\phi}$ , as described above.

# Score or Rao test

## Rao-score test

Let  $\hat{\beta} = (\hat{\beta}_A, \hat{\beta}_B)$  be the **unrestricted** maximum likelihood estimate and let  $\hat{\beta}_0 = (\hat{\beta}_{A,0}, \beta_0)$  be the **restricted** maximum likelihood estimate. Moreover, let

$$\ell_B(\beta, \phi) = \frac{\partial}{\partial \beta_B} \ell(\beta, \phi),$$

namely the **block** of the score function associated with  $\beta_B$ . The quantity

$$W_u = \ell_B(\hat{\beta}_0, \hat{\phi})^T \widehat{\text{var}}(\hat{\beta}_B) \ell_B(\hat{\beta}_0, \hat{\phi}),$$

is called **Rao-score test**. Here  $\widehat{\text{var}}(\hat{\beta}_B)$  is the appropriate **block** of  $(\mathbf{X}^T \tilde{\mathbf{W}} \mathbf{X})^{-1}$  where  $\tilde{\mathbf{W}}$  is estimated using the **restricted**  $\hat{\beta}_0$ . Under  $H_0$  this quantity is **approximately** distributed as

$$W_u \dot{\sim} \chi_q^2,$$

a  $\chi^2$  distribution with  $q$  **degrees of freedom**. The **p-value** is  $\text{Pr}( > \text{Chi} ) = \mathbb{P}(W_u > w_{u, \text{obs}})$ .

The Rao-score test is arguably the **least common**. When  $\phi$  is unknown, there are several variants depending on how it is estimated.

## A graphical representation when $p = 1$

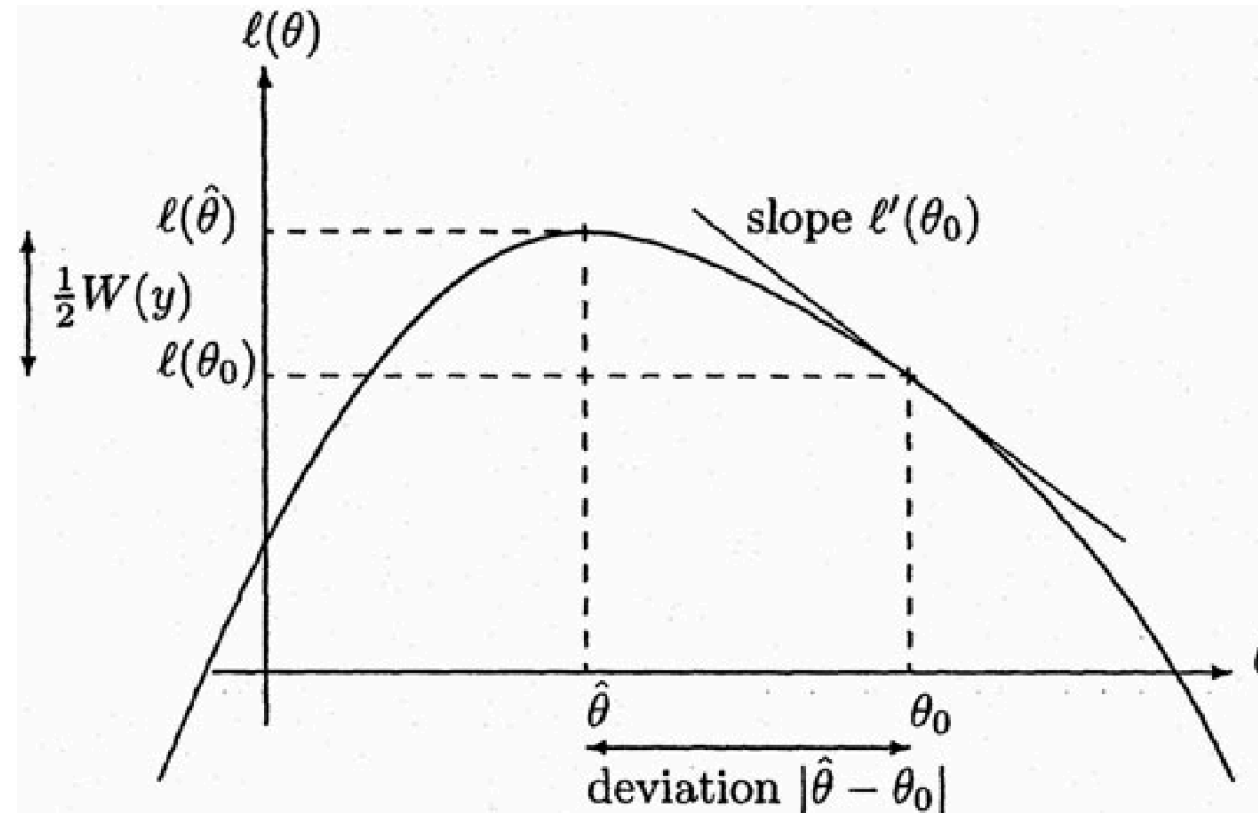


Figure 4.2 *Three test functions connected to the log-likelihood*

- Figure taken from Azzalini (2008). This is also the **cover** of the **book**!

## Three asymptotically equivalent tests

The Wald test, the Score test and the log-likelihood ratio test are **asymptotically equivalent**, that is, these tests give the **same number** for large values of  $n$ . We have that

$$W_e = W + o_p(1), \quad W_u = W + o_p(1),$$

where  $o_p(1)$  is a quantity that goes to 0 in probability as  $n \rightarrow \infty$ .

- When  $q = 1$ , we can also **invert**  $W_e$ ,  $W_u$  and  $W$  tests over  $\beta_0$  to obtain the corresponding **confidence interval**. This is often done **numerically** for  $W_u$  and  $W$ .
  - The Wald test depends on the parametrization. When considering a transformation of  $\beta$ , the variance must be adjusted using the derivative of the transformation (**delta method**).<sup>1</sup>
  - On the other hand, both the LRT and the score are **invariant**, and therefore we can simply **transform** the **extremes** of the **original interval** without further corrections.
1. Transforming the extremes of the Wald confidence interval “works” in the sense that it produces a valid confidence interval, but it is **not** the Wald interval in the transformed scale.

## Comparison with the Gaussian linear model

In the Gaussian linear model all tests are **equivalent** if  $\phi = \sigma^2$  is **known**. We have

$$W = W_e = W_u = \frac{\|\mathbf{Y} - \mathbf{X}\hat{\beta}_0\|^2 - \|\mathbf{Y} - \mathbf{X}\hat{\beta}\|^2}{\sigma^2} \sim \chi_q^2.$$

The  $\chi_q^2$  distribution is **exact** and not an approximation thanks to Cochran's theorem.

- Consider the **log-likelihood ratio** for testing  $H_0 : \beta_B = \beta_0$ . Suppose  $\sigma^2$  is **unknown**, then:

$$\begin{aligned} W &= 2[\ell(\hat{\beta}, \hat{\phi}) - \ell(\hat{\beta}_0, \hat{\phi})] = \frac{\|\mathbf{Y} - \mathbf{X}\hat{\beta}_0\|^2 - \|\mathbf{Y} - \mathbf{X}\hat{\beta}\|^2}{\hat{\phi}} = q \frac{(\|\mathbf{Y} - \mathbf{X}\hat{\beta}_0\|^2 - \|\mathbf{Y} - \mathbf{X}\hat{\beta}\|^2)/q}{\|\mathbf{Y} - \mathbf{X}\hat{\beta}\|^2/(n-p)} \\ &= qF, \end{aligned}$$

where  $F \sim F_{q,n-p}$  is the usual **Snedecor's** F. Indeed  $qF$  is **approximately** distributed as  $\chi_q^2$  for large values of  $n$ .

- The quantities  $W_e$ ,  $W_u$ , and  $W$  are the natural extension of the F-statistic for GLMs. They are **approximately** distributed as  $\chi_q^2$  with  $q$  **degrees of freedom**.

## Example: Beetles data

- We would like to use the **Wald**, the **Rao-score** and the **log-likelihood ratio** tests to verify the hypothesis  $H_0 : \beta_2 = 0$ , that is the relevance of **logdose** in predicting the response.
- In this case, we have  $q = 1$  (Df) because there is only one parameter under scrutiny.

| Test for the hypothesis $H_0 : \beta_2 = 0$ | Chi     | Df | Pr(>Chi)    |
|---------------------------------------------|---------|----|-------------|
| $W_e$ - <b>Wald test</b>                    | 138.488 | 1  | $\approx 0$ |
| $W_u$ - <b>Rao-score test</b>               | 227.580 | 1  | $\approx 0$ |
| $W$ - <b>Log-likelihood ratio test</b>      | 272.970 | 1  | $\approx 0$ |

- As one may expect, the test values are not identical. Here the sample size is  $n = 8$ , which is definitely not a big number, therefore we are far from the **asymptotic regime**.
- However, the practical conclusions are identical. All tests strongly **reject** the null hypothesis.
- We previously obtained the **Wald statistic**  $z_j$  and indeed  $z_j^2 = 11.76811^2 = 138.488 = W_e$ .

## Example: Beetles data

- Any statistical test can be **inverted**, namely we find all the values  $\beta_0$  such that we do **not reject the null hypothesis**. This generates a **confidence interval**.
- For the Wald test, the inversion is done analytically, producing the “usual” confidence interval.
- For the Rao-score and the log-likelihood ratio we need numerical procedures.
- In the **Beetles** data, the three tests produce the following confidence intervals for  $\beta_2$ , associated to **logdose**.

| Confidence intervals for $\beta_2$ at a 95% level | 2.5%   | 97.5%  |
|---------------------------------------------------|--------|--------|
| $W_e$ - <b>Wald test</b>                          | 28.563 | 39.978 |
| $W_u$ - <b>Rao-score test</b>                     | 28.588 | 39.957 |
| $W$ - <b>Log-likelihood ratio test</b>            | 28.854 | 40.301 |

- The Wald interval was also computed before. The three tests produce **nearly identical** intervals.
- Wald is always symmetric around  $\hat{\beta}_j$ , whereas Rao and the log-likelihood ratio are typically **asymmetric**, depending on the shape of the likelihood function.

## Example: Aids data

- Let us now perform the same analysis for the Aids data. Again, we test the null hypothesis  $H_0 : \beta_2 = 0$ , which is the relevance of period in predicting the response.

| Test for the hypothesis $H_0 : \beta_2 = 0$ | Chi     | Df | Pr(>Chi)    |
|---------------------------------------------|---------|----|-------------|
| $W_e$ - Wald test                           | 135.602 | 1  | $\approx 0$ |
| $W_u$ - Rao-score test                      | 163.586 | 1  | $\approx 0$ |
| $W$ - Log-likelihood ratio test             | 178.551 | 1  | $\approx 0$ |

- As before, despite their numerical differences, all the tests reject the null hypothesis. We previously obtained the Wald statistic  $z_j = 11.645$  and indeed  $z_j^2 = 11.645^2 = 135.6 = W_e$ .

| Confidence intervals for $\beta_2$ at a 95% level | 2.5%   | 97.5%  |
|---------------------------------------------------|--------|--------|
| $W_e$ - Wald test                                 | 0.2154 | 0.3025 |
| $W_u$ - Rao-score test                            | 0.2155 | 0.3025 |
| $W$ - Log-likelihood ratio test                   | 0.2165 | 0.3037 |

## Example: Aids data

- We are actually interested in a confidence interval for the quantity  $100 \times (\exp(\beta_2) - 1)$ , which is the **percentage increase** of **deaths** after each period.
- Thanks to the **invariance property** of the Rao-score and the log-likelihood ratio tests, we can simply **transform** the **original intervals** for  $\beta_2$ .
- If the extremes of the log-likelihood ratio interval are  $C_{\text{low}}, C_{\text{high}}$ , then the new interval is

$$[100 \times (\exp(C_{\text{low}}) - 1), 100 \times (\exp(C_{\text{high}}) - 1)].$$

and similarly for the Rao-score case. These are reported below.

| Confidence intervals for $100[\exp(\beta_2) - 1]$ at a 95% level | 2.5%  | 97.5% |
|------------------------------------------------------------------|-------|-------|
| $W_u$ - <b>Rao-score test</b>                                    | 24.04 | 35.32 |
| $W$ - <b>Log-likelihood ratio test</b>                           | 24.17 | 35.49 |

- The **average** percentage increase is between 24% and 35% each **period**, with a 95% **confidence**.
- These confidence intervals are always **positive**, which is desirable because they are percentages.

## Example: Aids data

- In the Wald case, we cannot simply transform the extremes of the intervals. Indeed, that would lead to a **valid** confidence interval that is no longer of Wald type (*Lo sbagliato* 🍷).
- Instead, we first need to **adjust** the **variance** according to the **delta method**, obtaining

$$\widehat{\text{var}}\{100[\exp(\hat{\beta}_2) - 1]\} = 100^2 \exp(2\hat{\beta}_2) \text{var}(\hat{\beta}_2) = 8.301184.$$

- The Wald confidence interval for  $100[\exp(\hat{\beta}_2) - 1]$  therefore is

$$100[\exp(\hat{\beta}_2) - 1] \pm z_{1-\alpha/2} \widehat{\text{se}}\{100[\exp(\hat{\beta}_2) - 1]\}.$$

| Confidence intervals for $100[\exp(\beta_2) - 1]$ at a 95% level | 2.5%  | 97.5% |
|------------------------------------------------------------------|-------|-------|
| $W_e$ - <b>Wald test</b>                                         | 23.91 | 35.21 |
| " <i>Lo sbagliato</i> " - transformed Wald                       | 24.03 | 35.33 |

Whenever there are restrictions on the parametric space, as in this case, Wald is typically **problematic**. Here, it **could** lead to **negative values**, which is absurd.

# Deviance, model checking, residuals

## Deviance: some intuitions

- In a Gaussian linear model, we called **deviance** the residual sum of squares, that is

$$D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = \sum_{i=1}^n (y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}})^2 = \sum_{i=1}^n (y_i - \hat{\mu}_i)^2.$$

- The residual sum of squares  $D(\mathbf{y}; \hat{\boldsymbol{\mu}})$  is a **goodness of fit** measure. The lower the deviance, the higher the quality of the predictions.
- When  $\sigma^2$  is **known**, the distribution of the **scaled deviance** is

$$\frac{D(\mathbf{Y}; \hat{\boldsymbol{\mu}})}{\sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^n (Y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}})^2 \sim \chi_{n-p}^2.$$

- When  $\sigma^2$  is known, the difference of scaled deviances of two nested models is:

$$W = \frac{D(\mathbf{Y}; \hat{\boldsymbol{\mu}}_0) - D(\mathbf{Y}; \hat{\boldsymbol{\mu}})}{\sigma^2} = \frac{\|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_0\|^2 - \|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2}{\sigma^2} \sim \chi_q^2.$$

- The natural question is: what is a natural **generalization** of the deviance for GLMs?

## Example: Beetles data, saturated model

- Let us consider again the **Beetles** data and the predictions  $\hat{\mu}_i$ , based on  $p = 2$  parameters. These predictions are not perfect but that may be due to chance.

| $m_i$ | <b>deaths</b> ( $S_i$ ) | <b>logdose</b> ( $x_i$ ) | $Y_i = S_i/m_i$ | $\hat{\mu}_i$ |
|-------|-------------------------|--------------------------|-----------------|---------------|
| 59    | 6                       | 1.691                    | 0.102           | 0.059         |
| 60    | 13                      | 1.724                    | 0.217           | 0.164         |
| 62    | 18                      | 1.755                    | 0.290           | 0.362         |
| 56    | 28                      | 1.784                    | 0.500           | 0.605         |
| 63    | 52                      | 1.811                    | 0.825           | 0.795         |
| 59    | 53                      | 1.837                    | 0.898           | 0.903         |
| 62    | 61                      | 1.861                    | 0.984           | 0.955         |
| 60    | 60                      | 1.884                    | 1.000           | 0.979         |

- The **empirical proportions**  $s_i/m_i$  can be seen as **estimates** of the **most flexible model**, in which every observation  $Y_i$  has its own mean  $\mu_i$ . We call it the **saturated model** because  $p = n$ .

# Saturated model

- Let us express the **log-likelihood** of a GLM as a function of the mean  $\boldsymbol{\mu} = (\mu_1, \dots, \mu_n)^T$ .
- When evaluated in the maximum likelihood, this gives:

$$\ell_{\mathcal{M}}(\hat{\boldsymbol{\mu}}, \phi) = \sum_{i=1}^n \left\{ \omega_i \frac{y_i \theta(\hat{\mu}_i) - b(\theta(\hat{\mu}_i))}{\phi} + c(y_i, \phi) \right\}.$$

The maximum likelihood for each  $\mu_i$  is **restricted**, in the sense that it depends on the  $p$  parameters of the **linear predictor**  $\mathbf{x}_i^T \boldsymbol{\beta}$  through the link function  $g(\mu_i) = \mathbf{x}_i^T \boldsymbol{\beta}$ .

- In the **saturated model** the means  $\mu_i$  are **unrestricted**: each parameter is estimated separately, giving the maximum likelihood estimate  $\hat{\mu}_{i,\text{sat}} = y_i$ . This happens whenever  $p = n$ .
- When evaluated in the maximum, the log-likelihood of the saturated model is

$$\ell_{\mathcal{M}}(\mathbf{y}, \phi) = \sum_{i=1}^n \left\{ \omega_i \frac{y_i \theta(y_i) - b(\theta(y_i))}{\phi} + c(y_i, \phi) \right\}.$$

- The saturated model is the **most complex model** we can think of.

# Deviance

The **deviance** of a GLM is defined as

$$\begin{aligned} D(\mathbf{y}; \hat{\boldsymbol{\mu}}) &:= \phi W = \phi 2[\ell_{\mathcal{M}}(\mathbf{y}, \phi) - \ell_{\mathcal{M}}(\hat{\boldsymbol{\mu}}, \phi)] \\ &= 2 \sum_{i=1}^n \omega_i \{y_i[\theta(y_i) - \theta(\hat{\mu}_i)] - [b(\theta(y_i)) - b(\theta(\hat{\mu}_i))]\}. \end{aligned}$$

The quantity  $D(\mathbf{y}; \hat{\boldsymbol{\mu}})/\phi$  is called **scaled deviance** and it corresponds to a **log-likelihood ratio test**  $W$  in which the current model is tested against the saturated model.

- By definition, the deviance is **positive**:  $D(\mathbf{y}; \hat{\boldsymbol{\mu}}) \geq 0$ , because  $\ell_{\mathcal{M}}(\mathbf{y}, \phi) \geq \ell_{\mathcal{M}}(\hat{\boldsymbol{\mu}}, \phi)$ .
- The deviance of the saturated model is  $D(\mathbf{y}; \mathbf{y}) = 0$ .
- The deviance describes a **lack of fit**: the higher the deviance, the poorer the fit.
- It measures the discrepancy between the saturated model and a model using  $p < n$  parameters.
- The deviance is a function of  $\boldsymbol{\mu}$ , therefore its definition does not depend on the link function  $g(\cdot)$ .

## Deviance and log-likelihood ratio test

Let us consider two **nested models**  $M_0 \subset M_1$ . The **reduced model**  $M_0$  has  $p_0$  parameters and predictions  $\hat{\mu}_0$ . The **full model**  $M_1$  has  $p$  parameters  $\hat{\mu}_1$ .

The **log-likelihood ratio test**  $W$  for testing model  $M_0$  against model  $M_1$ <sup>1</sup> can be written as

$$W = 2[\ell_{\mathcal{M}}(\hat{\mu}, \hat{\phi}) - \ell_{\mathcal{M}}(\hat{\mu}_0, \hat{\phi})] = \frac{D(\mathbf{Y}; \hat{\mu}_0) - D(\mathbf{Y}; \hat{\mu})}{\hat{\phi}} \sim \chi_q^2.$$

where  $q = p - p_0$  are the degrees of freedom.

- The log-likelihood ratio can be interpreted as a difference of scaled deviances. This explains why it is popular in GLMs for comparing nested models.
  - There is also a strong parallelism with the Gaussian linear model.
1. More formally, we should say that we are testing the hypothesis  $H_0 : \beta_B = \mathbf{0}$  against the alternative  $H_1 : \beta_B \neq \mathbf{0}$ . I hope you can tolerate this slight linguistic abuse.

# The null model

Let us consider a model  $M_{\text{null}}$  with **no covariates** and one parameter ( $p = 1$ ), i.e. the **intercept**.  
The predicted values are all equal to

$$\hat{\mu}_{\text{null}} = (g^{-1}(\hat{\beta}_1), \dots, g^{-1}(\hat{\beta}_1))^T.$$

We call  $M_{\text{null}}$  the **null model** and  $D(\mathbf{y}; \hat{\mu}_{\text{null}})$  the **null deviance**.

- The null model is the “opposite” of the saturated model. It is the **simplest** among all models and the one having the **highest deviance**.
- Indeed, the following inequalities hold:

$$0 = D(\mathbf{y}; \mathbf{y}) \leq D(\mathbf{y}; \hat{\mu}) \leq D(\mathbf{y}; \hat{\mu}_{\text{null}}).$$

- It is sometimes useful to test the current model against the null model:

$$W = \frac{D(\mathbf{Y}; \hat{\mu}_{\text{null}}) - D(\mathbf{Y}; \hat{\mu})}{\hat{\phi}} \sim \chi_{p-1}^2.$$

If  $H_0$  is not rejected, it means that all the **covariates** are regarded as **irrelevant**.

## Pearson $X^2$ statistic

- The deviance is a **log-likelihood ratio** test between a given model and the saturated model, rescaled by  $\phi$ .
- Hence, we may consider another test, like the **Rao-Score**, to obtain an alternative definition.

Let  $W_u$  be the **Rao-Score** test comparing model  $M$  with the **saturated model**. Then, it holds:

$$\phi W_u = X^2 = \sum_{i=1}^n \omega_i \frac{(y_i - \hat{\mu}_i)^2}{v(\hat{\mu}_i)},$$

which is known as generalized **Pearson chi-squared statistic**.

- Karl Pearson introduced  $X^2$  in 1900 for testing various hypotheses using the chi-squared distribution, such as the hypothesis of independence in contingency tables.
- Since  $W_u$  and  $W$  are **asymptotically equivalent**, so will be the chi-squared statistic  $X^2$  and the deviance  $D(\mathbf{Y}; \hat{\boldsymbol{\mu}})$  for large values of  $n$ .

# Deviance of a Gaussian linear model

- In the classical **Gaussian linear model**, we have that  $\theta_i = \mu_i$ ,  $b(\theta_i) = \theta_i^2/2$  and  $\omega_i = 1$ . Thus  $\theta(y_i) = y_i$  and

$$\begin{aligned} D(\mathbf{y}; \hat{\boldsymbol{\mu}}) &= 2 \sum_{i=1}^n \omega_i \{y_i [\theta(y_i) - \theta(\hat{\mu}_i)] - [b(\theta(y_i)) - b(\theta(\hat{\mu}_i))]\} \\ &= 2 \sum_{i=1}^n \{y_i(y_i - \hat{\mu}_i) - y_i^2/2 + \hat{\mu}_i^2/2\} \\ &= \sum_{i=1}^n (y_i^2 - 2y_i\hat{\mu}_i + \hat{\mu}_i^2) = \sum_{i=1}^n (y_i - \hat{\mu}_i)^2. \end{aligned}$$

- In the Gaussian case, the deviance is the **residual sum of squares** and  $D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = X^2$ .
- Note that the **null deviance** is obtained with  $\hat{\boldsymbol{\mu}}_{\text{null}} = (\bar{y}, \dots, \bar{y})$  so that

$$D(\mathbf{y}; \hat{\boldsymbol{\mu}}_{\text{null}}) = \sum_{i=1}^n (y_i - \bar{y})^2,$$

namely the so-called **total deviance**.

## Deviance of a Poisson model

- Let us consider a **Poisson regression** model, that is  $\theta_i = \log \mu_i$  and  $b(\theta_i) = \exp(\theta_i) = \mu_i$ . Then  $\theta(y_i) = \log y_i$  and

$$\begin{aligned} D(\mathbf{y}; \hat{\boldsymbol{\mu}}) &= 2 \sum_{i=1}^n \{y_i(\log y_i - \log \hat{\mu}_i) - y_i + \hat{\mu}_i\} \\ &= 2 \sum_{i=1}^n \{y_i \log(y_i / \hat{\mu}_i) - y_i + \hat{\mu}_i\}, \end{aligned}$$

with the **convention** that  $y_i \log(y_i / \hat{\mu}_i) = 0$  whenever  $y_i = 0$ .

- The  $X^2$  statistic in this case has a very simple form

$$X^2 = \sum_{i=1}^n \frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i} = \sum_{i=1}^n \frac{(\text{observed}_i - \text{fitted}_i)^2}{\text{fitted}_i}.$$

As discussed in Salvan et al. (2020), Example 2.12, this can be seen as a **quadratic approximation** of the **deviance**, which is valid for large values of  $n$ .

# Deviance of a binomial model I

- Let us consider a **Binomial regression** model  $m_i Y_i = S_i \sim \text{Binomial}(m_i, \pi_i)$  with  $\mu_i = \pi_i$ . Then, up to an **additive constant** not depending on  $\mu$ ,

$$\ell_{\mathcal{M}}(\hat{\mu}) = \sum_{i=1}^n \{m_i y_i \log(\hat{\mu}_i) + m_i(1 - y_i) \log(1 - \hat{\mu}_i)\}.$$

Therefore, under the convention  $x \log(x) = 0$  as before, the deviance is

$$\begin{aligned} D(\mathbf{y}; \hat{\mu}) &= 2 \sum_{i=1}^n m_i \left\{ y_i \log \left( \frac{y_i}{\hat{\mu}_i} \right) + (1 - y_i) \log \left( \frac{1 - y_i}{1 - \hat{\mu}_i} \right) \right\} \\ &= 2 \sum_{i=1}^n \left\{ m_i y_i \log \left( \frac{m_i y_i}{m_i \hat{\mu}_i} \right) + (m_i - m_i y_i) \log \left( \frac{m_i - m_i y_i}{m_i - m_i \hat{\mu}_i} \right) \right\}. \end{aligned}$$

- The quantities  $m_i y_i$  and  $m_i - m_i y_i$  can be interpreted as the number of observed **successes** and **failures**, respectively. Similarly,  $m_i \hat{\mu}_i$  and  $m_i - m_i \hat{\mu}_i$  represent their **predictions**. Hence, we can write

$$D(\mathbf{y}; \hat{\mu}) = 2 \sum_{j=1}^{2n} \text{observed}_j \log \left( \frac{\text{observed}_j}{\text{fitted}_j} \right)$$

## Deviance of a binomial model II

- The  $X^2$  statistic of a binomial model, recalling that  $v(\mu_i) = \mu_i(1 - \mu_i)$ , equals to

$$\begin{aligned} X^2 &= \sum_{i=1}^n \frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i(1 - \hat{\mu}_i)/m_i} \\ &= \sum_{i=1}^n \frac{(m_i y_i - m_i \hat{\mu}_i)^2}{m_i \hat{\mu}_i} + \sum_{i=1}^n \frac{[(m_i - m_i y_i) - (m_i - m_i \hat{\mu}_i)]^2}{m_i - m_i \hat{\mu}_i}. \end{aligned}$$

The second representation follows after some **algebra**.

- The second equation shows that we can write

$$X^2 = \sum_{j=1}^{2n} \frac{(\text{observed}_j - \text{fitted}_j)^2}{\text{fitted}_j}.$$

- As already mentioned, the  $X^2$  statistic can be seen as a **quadratic approximation** of the **deviance**.

# Deviance as goodness of fit measure I

- The deviance is a useful **descriptive measure** of the **goodness of fit**.
- It is **tempting** to use the deviance as a formal **statistical test**, to verify if the current model is adequate compared to the saturated model.
- Suppose  $\phi$  were **known**, then in the Gaussian case we would have  $D(\mathbf{y}; \hat{\boldsymbol{\mu}})/\phi \sim \chi_{n-p}^2$ , which would allow us to check the adequacy of the model.
- Unfortunately, whenever  $\hat{\phi}$  is **estimated** we obtain  $D(\mathbf{y}; \hat{\boldsymbol{\mu}})/\hat{\phi} \approx n - p$ , and exactly  $n - p$  in the Gaussian case, so this strategy cannot be used.
- On the other hand, for example in Poisson and binomial regression, we have  $\phi = 1$ . Hence, intuition tells us that, at least approximately, we should have

$$D(\mathbf{y}; \hat{\boldsymbol{\mu}}) \dot{\sim} \chi_{n-p}^2.$$

- Unfortunately, this is **not** the **case**: the saturated model is an “irregular case” in that the number of parameters  $p = n$  grows with the sample size.
- The usual “large  $n$ ” approximation does not hold in general, e.g. because  $\chi_{n-p}^2$  itself depends on  $n$ .

## Deviance as goodness of fit measure II

- Despite this bad news, it turns out that in some special cases, the  $\chi^2_{n-p}$  approximation is still valid even for fixed values of  $n$ .

The **deviance**  $D(\mathbf{y}; \hat{\boldsymbol{\mu}})$  and the  $X^2$  **Pearson statistic** are approximately distributed as a  $\chi^2_{n-p}$  in the following cases:

- In binomial regression, for large values of  $m_i$  (small-dispersion asymptotics);
  - In Poisson regression, for large values of the estimated means  $\hat{\mu}_i$  (say  $\hat{\mu}_i \geq 5$ );
  - When  $\phi$  is known and  $\phi \rightarrow 0$  (small-dispersion asymptotics).
- **Small-dispersion asymptotics** describe an alternative limiting regime in which the variance of the observations tends to 0.
  - In binomial regression the  $X^2$  statistic converges to  $\chi^2_{n-p}$  more quickly than the deviance and has a **more trustworthy** p-value when some expected success or failure totals are less than about five.
  - The  $\chi^2_{n-p}$  approximation is **very poor** for binary regression, i.e. when  $m_i = 1$ .

## On pseudo- $R^2$

- There exist several generalizations of the  $R^2$  statistic for linear models, called **pseudo- $R^2$**  (e.g. McFadden, Cox & Snell, Nagelkerke, Tjur, etc.).
- These indices are **difficult to interpret** and could mislead those accustomed to the standard  $R^2$ . A pseudo- $R^2 \approx 0.4$  may indicate a nearly perfect fit (e.g. the **Beetles** data), which is confusing.
- On top of this, these pseudo- $R^2$  measures produce different answers depending on the **aggregation** of the data.
- The recommendation is to rely on indices tailored to the data at hand, such as the ROC curve for binary data, or the correlation between  $y$  and  $\hat{\mu}$ .
- The residual **deviance** is also a useful tool, especially for comparing models.
- Pseudo- $R^2$  measures are often shown by default in other software, such as SAS or SPSS.

## Example: Beetles data, output of summary

- This is what the **summary** of a GLM looks like. It is very similar to the **summary** of **lm**. At this stage of the course, you should be able to understand almost everything.

```
Call:
glm(formula = cbind(deaths, m - deaths) ~ logdose, family = "binomial",
     data = Beetles)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | -60.717  | 5.181      | -11.72  | <2e-16 *** |
| logdose     | 34.270   | 2.912      | 11.77   | <2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 284.202 on 7 degrees of freedom  
 Residual deviance: 11.232 on 6 degrees of freedom  
 AIC: 41.43

Number of Fisher Scoring iterations: 4

- **Null deviance** corresponds to the **null deviance**  $D(\mathbf{y}; \hat{\mu}_{\text{null}})$ .
- **Residual deviance** corresponds to the **deviance**  $D(\mathbf{y}; \hat{\mu})$  of the current model.

## Example: Beetles data, output of anova

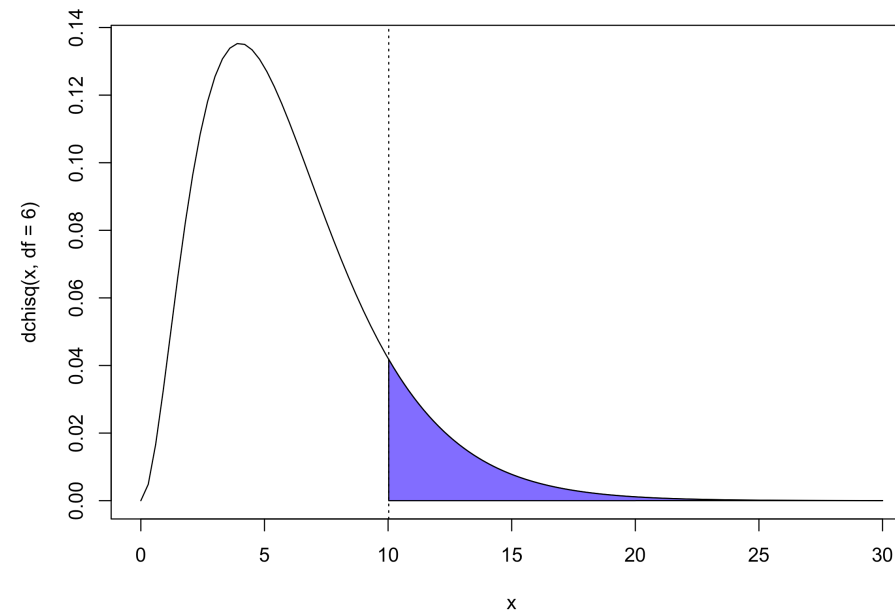
- `anova(model0, model1)` computes the **log-likelihood ratio test** comparing two nested models: the **reduced** model  $M_0$  with  $p_0$  parameters and the **full** model  $M_1$  with  $p$  parameters.

Analysis of Deviance Table

```
Model 1: cbind(deaths, m - deaths) ~ 1
Model 2: cbind(deaths, m - deaths) ~ logdose
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1         7    284.202
2         6     11.232  1    272.97 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- **Resid Df** are the **degrees of freedom** of the **deviances**, that is  $n - p_0$  and  $n - p$ , respectively.
- **Resid. Dev** are the **deviances** of the **reduced** model  $D(\mathbf{y}; \hat{\boldsymbol{\mu}}_0)$  and the **full** model  $D(\mathbf{y}; \hat{\boldsymbol{\mu}})$ . In this example, the reduced model is also the **null model**.
- **Df** refers to the **degrees of freedom**  $q = p - p_0$  of the test, which is  $q = 1$  in this case.
- **Deviance** indicates the **change in deviance**, that is  $\phi W = D(\mathbf{y}; \hat{\boldsymbol{\mu}}_0) - D(\mathbf{y}; \hat{\boldsymbol{\mu}})$ .
- **Pr(>Chi)** is the **p-value** of the log-likelihood ratio test  $W$ .

## Example: Beetles data, goodness of fit



- The **deviance** equals  $D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 11.232$ , with  $n - p = 8 - 2 = 6$  degrees of freedom. The observed  $X^2$  **Pearson statistic** equals 10.027.
- Using the  $X^2$  statistic with 6 degrees of freedom, we obtain the p-value  $\mathbb{P}(\chi_6^2 > 10.027) = 0.124$ , as pictured above. There is therefore **no evidence** of lack of fit: this model is **compatible** with the saturated one.

## Example: **Aids** data, output of **summary**

- The **summary** of the Poisson regression model for the **Aids** data is shown below.

```
Call:
glm(formula = deaths ~ period, family = "poisson", data = Aids)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.30365    0.25387   1.196   0.232
period       0.25896    0.02224  11.645  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 208.754  on 13  degrees of freedom
Residual deviance:  30.203  on 12  degrees of freedom
AIC: 86.949

Number of Fisher Scoring iterations: 5
```

## Example: **Aids** data, output of **anova** and **lrtest**

Analysis of Deviance Table

Model 1: deaths ~ 1

Model 2: deaths ~ period

|   | Resid. Df | Resid. Dev | Df | Deviance | Pr(>Chi)      |
|---|-----------|------------|----|----------|---------------|
| 1 | 13        | 208.754    |    |          |               |
| 2 | 12        | 30.203     | 1  | 178.55   | < 2.2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

The use of the term **deviance** to indicate the **difference** between two deviances is a bit **misleading**. I do not know the reason, but the **lrtest** function of the **lmtest** package changed it.

Likelihood ratio test

Model 1: deaths ~ 1

Model 2: deaths ~ period

|   | #Df | LogLik   | Df | Chisq  | Pr(>Chisq)    |
|---|-----|----------|----|--------|---------------|
| 1 | 1   | -130.750 |    |        |               |
| 2 | 2   | -41.475  | 1  | 178.55 | < 2.2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

## Example: Aids data, goodness of fit

- The **deviance** equals  $D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 30.203$ , with  $n - p = 14 - 2 = 12$  degrees of freedom. The observed  $X^2$  **Pearson statistic** equals 29.92.
- Using the  $X^2$  statistic with 12 degrees of freedom, we obtain the p-value  $\mathbb{P}(\chi_{12}^2 > 29.92) = 0.0028$ , therefore **rejecting** the hypothesis that this model has a fit comparable to the saturated one.
- The  $X^2$  goodness of fit test indicates a potential issue with the model, but it does not explain **why**.
- A few remedies could be:
  - a. Choosing a different **link function**;
  - b. Including an **additional covariate** (if available), and/or considering non-linear **transformations** of the available ones;
  - c. Choosing a different **distribution** instead of the Poisson, such as the negative binomial;
  - d. Accounting for **overdispersion** using quasi-likelihoods, that is, estimating  $\phi$  from the data rather than fixing it to  $\phi = 1$ .
- It turns out that selecting the link function  $g(\mu_i) = \sqrt{\mu_i}$  yields a much better fit with  $X^2 = 17.09$ , whose p-value is 0.146. This is not the only possible solution.

# Residuals

- Linear models have an additive structure  $y_i = \mathbf{x}_i^T \beta + \epsilon_i$  therefore the residuals can be estimated as

$$r_i = y_i - \hat{\mu}_i.$$

We call these the **response residuals**.

- GLMs do not have an additive decomposition, therefore we need to define a good **generalization** of **residuals**. There are at least two alternatives: Pearson and deviance residuals.
- Ideally, we would like residuals to have approximately 0 mean and unitary variance, but these properties will not hold exactly.
- The analysis of the residuals is very helpful for identifying any misspecification, as well as for hinting at the solution. In particular, it is useful for instance for:
  - a. Choosing the correct variance function  $v(\mu_i)$ , i.e. the correct response distribution;
  - b. Choosing the correct link function;
  - c. Identifying latent patterns, often an indication of an omitted variable;
  - d. Identifying potential outliers and leverage points.

## Pearson residuals

For a GLM with variance function  $v(\mu_i)$ , we call **Pearson residuals** the following quantities:

$$r_{i,P} = \frac{y_i - \hat{\mu}_i}{\sqrt{v(\hat{\mu}_i)/\omega_i}}, \quad i = 1, \dots, n.$$

- Pearson residuals rescale the response residuals, accounting for **heteroscedasticity**.
- Pearson residuals owe their name to the fact that the  $X^2$  statistic is obtained as

$$X^2 = \sum_{i=1}^n r_{i,P}^2 = \sum_{i=1}^n \omega_i \frac{(y_i - \hat{\mu}_i)^2}{v(\hat{\mu}_i)}.$$

Moreover, the dispersion parameter, when present, can be estimated as

$$\hat{\phi} = \frac{1}{n - p} \sum_{i=1}^n r_{i,P}^2.$$

## Deviance residuals

The deviance of a GLM can be written as a sum of individual contributions  $D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = \sum_{i=1}^n d_i$  and

$$d_i = 2\omega_i \{y_i[\theta(y_i) - \theta(\hat{\mu}_i)] - [b(\theta(y_i)) - b(\theta(\hat{\mu}_i))]\}, \quad i = 1, \dots, n.$$

We call **deviance residuals** the following quantities:

$$r_{i,D} = \text{sign}(y_i - \hat{\mu}_i) \sqrt{d_i}, \quad i = 1, \dots, n.$$

- By definition, the deviance is obtained as

$$D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = \sum_{i=1}^n r_{i,D}^2.$$

- Deviance residuals are the **default** choice in the **residuals** **R** function.
- Pearson and deviance residuals are **asymptotically equivalent**, therefore these two quantities are often very similar in practice.

## A weighted projection matrix

- In linear models, we considered the **hat matrix**  $\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$ , whose diagonal elements  $h_i$  are called **leverages** and it holds  $\text{var}(r_i) = \sigma^2(1 - h_i)$ .

- Several arguments can be invoked to justify the following **weighted hat matrix** in GLMs

$$\mathbf{H}_W = \mathbf{W}^{1/2} \mathbf{X} (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W}^{1/2}.$$

This matrix is **symmetric** ( $\mathbf{H}_W = \mathbf{H}_W^T$ ) and **idempotent** ( $\mathbf{H}_W^2 = \mathbf{H}_W$ ), i.e. a projection matrix.

- We denote with  $h_{i,W}$  the diagonal elements of  $\mathbf{H}_W$ , which are the **leverages** of a GLM. In practice  $\mathbf{W}$  is estimated from the data, therefore the leverages will depend on the **response**.
- It can be shown, as in Agresti (2015), Section 4.4.5, that

$$\text{var}(y_i - \hat{\mu}_i) \approx (\phi/\omega_i)v(\mu_i)(1 - h_{i,W}) \implies \text{var}(r_{i,P}) \approx \phi(1 - h_{i,W}).$$

- A technical but deeper discussion about  $\mathbf{H}_W$  can be found in the Appendix of Chapter 4 of Agresti (2015), where stronger analogies with linear models are shown.

## Standardized residuals

- In analogy with what has been done for linear models, we can consider the **standardized** version of Pearson and deviance residuals.
- **Standardized Pearson residuals** are defined as

$$\tilde{r}_{i,P} = \frac{r_{i,P}}{\sqrt{\hat{\phi}(1 - \hat{h}_{i,W})}} = \frac{y_i - \hat{\mu}_i}{\sqrt{(\hat{\phi}/\omega_i)v(\hat{\mu}_i)(1 - \hat{h}_{i,W})}}, \quad i = 1, \dots, n,$$

where  $\hat{\phi}$  is an estimate of  $\phi$  (if unknown) and  $\hat{h}_{i,W}$  is an estimate of the leverages.

- **Standardized deviance residuals** are defined as

$$\tilde{r}_{i,D} = \frac{r_{i,D}}{\sqrt{\hat{\phi}(1 - \hat{h}_{i,W})}}, \quad i = 1, \dots, n.$$

- We can also obtain an **approximate Cook's distance** by considering

$$c_i = \tilde{r}_{i,P}^2 \frac{\hat{h}_{i,W}}{p(1 - \hat{h}_{i,W})}, \quad i = 1, \dots, n.$$

## On Q-Q plots and other practicalities

- It is sometimes recommended to check the normality of the Pearson/deviance residuals using Q-Q plots. Such a plot is also provided in **R**.
- Indeed, for example under small dispersion asymptotics or other specific conditions, Pearson residuals are **approximately** Gaussian.
- However, these conditions are often not met. For example, in binary data the response  $y_i \in \{0, 1\}$  can only assume two values and the residuals will not be normally distributed, even for large  $n$ .
- Actually, the analysis of the residuals in **binary data** does **not provide useful information**; see e.g. Salvan et al. (2020), Section 3.6.

The analysis of residuals in GLMs is often useful but should be taken *cum grano salis*, particularly when dealing with discrete responses that take only a few distinct values.

Overreliance on residual analysis can encourage automatic decisions rather than thoughtful, critical judgment. A good model should not be discarded solely on the basis of a “bad” diagnostic plot.

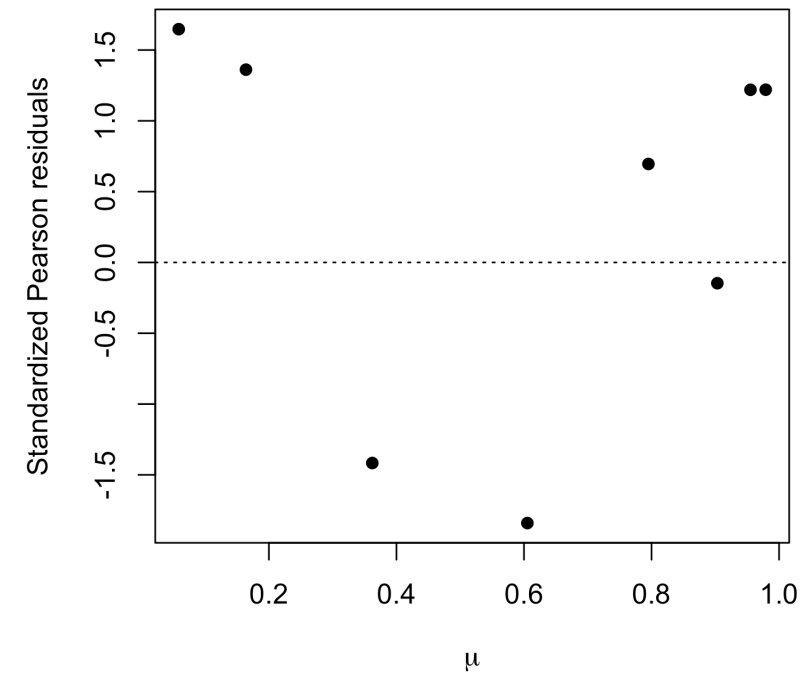
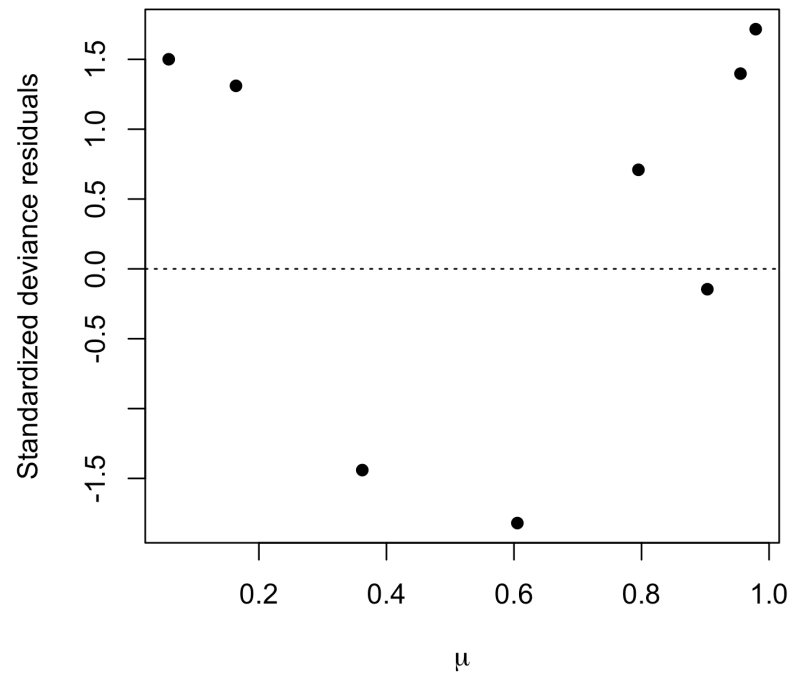
## On identifying and removing outliers

- The analysis of the residuals can also help in identifying outliers and influential points. However, we must be careful in drawing conclusions.
- An outlier might be detected as such simply as a consequence of **model misspecification**, e.g. an omitted variable.
- In the vast majority of cases, the presence of outliers should be dealt with by carefully modifying the model.
- There are instances in which outliers are actually **contaminated data points** (e.g. **age = -3**). If there is strong and contextual evidence that this might be the case, then these points should be **removed**. Otherwise, removing data points is a **bad practice**.

In statistical modelling, we wish to find a model that fits our data, not a dataset that is aligned with our prescribed model.

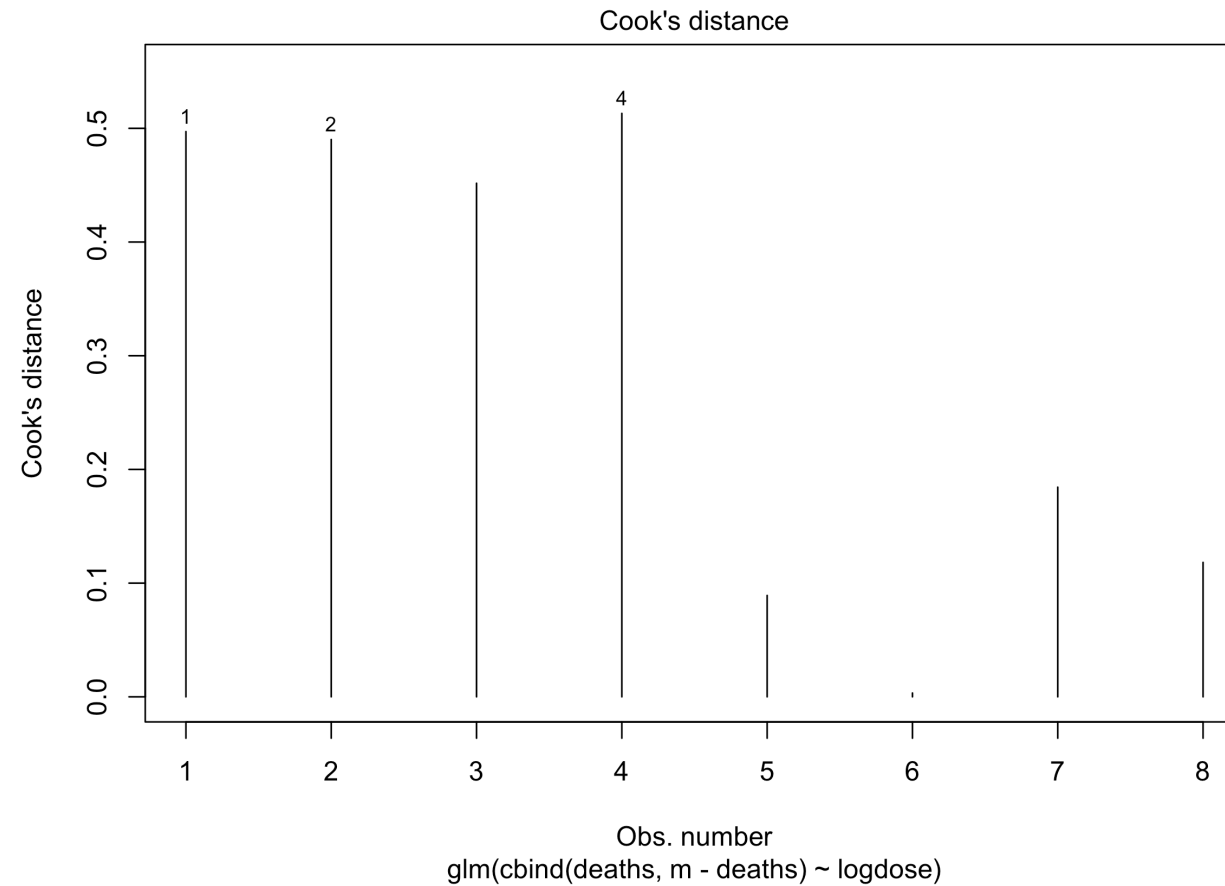
Discarding observations until the hypotheses are reasonable skews the overall analysis and does not answer any meaningful scientific or business question.

## Example: Beetles data



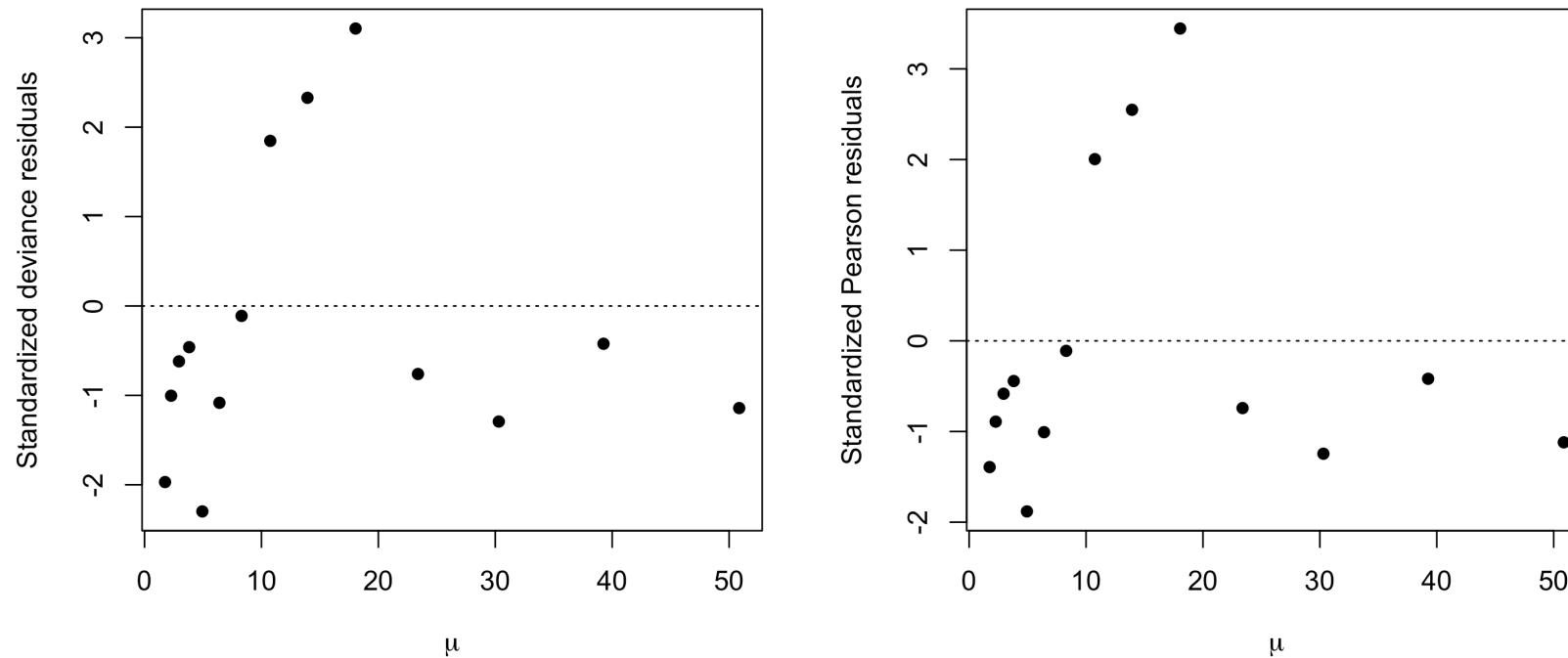
- The goodness of fit test did not detect any **noticeable difference** between this model and the saturated one. The analysis of the residuals confirms it.
- Deviance residuals and Pearson residuals are very similar, as expected.

## Example: Beetles data



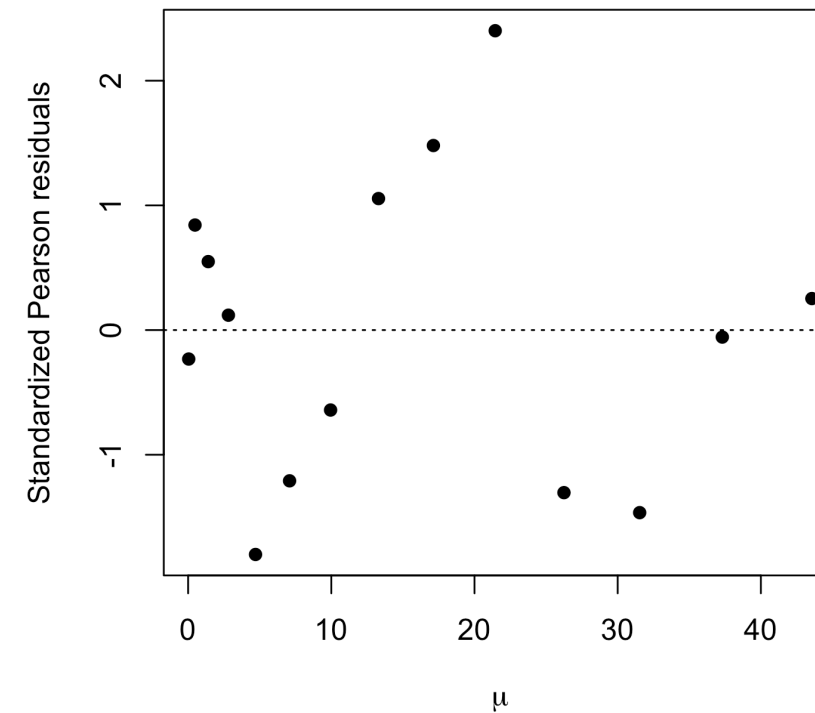
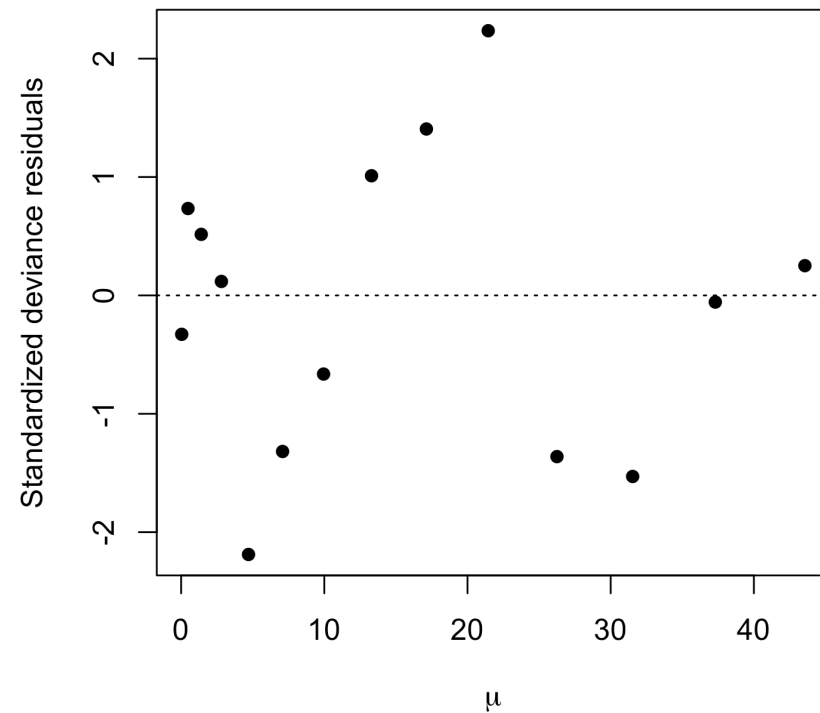
- The Cook's distance also confirms that there are no strong influential points.

## Example: Aids data



- From both residual plots it is evident that three observations are severely underestimated, while the others are slightly overestimated.
- The lack of fit can be remedied, in this case, by using a different link function.

## Example: Aids data



- We estimated a Poisson regression model with a **non-canonical link** function  $g(\mu_i) = \sqrt{\mu_i}$ .
- This yields a much **better fit**, as we previously discussed. The residuals are also better behaved.

# Model selection

# Model selection process

- **Model selection** for GLMs faces the same issues as for linear models.
- The selection process becomes more difficult as the number of explanatory variables  $p$  increases, because of the growth in possible effects and interactions. There are two competing goals:
  - a. The model should be complex enough to **fit the data well**;
  - b. On the other hand, it should **smooth rather than overfit** the data and ideally remain **simple to interpret**.
- Most research studies are designed to answer certain questions, which guide the choice.
- **Confirmatory analyses** use a restricted set of models, e.g. for testing a study hypothesis about an effect by comparing models **with and without** that effect.
- **Exploratory studies**, instead, search among possible models which may provide clues about the structure of effects or can raise questions for future research.
- In either case, it is helpful first to study the **marginal effect** of each predictor. Use **descriptive statistics** and a scatterplot matrix to get a feel for those effects.

# Automatic model selection

- With  $p$  **explanatory variables**, the number of potential models is the huge number

$$\sum_{k=0}^p \binom{p}{k} = 2^p.$$

- **Best subset selection** identifies the model that optimizes an **information criterion** e.g. AIC or BIC, which are defined as

$$\text{AIC} = -2\ell(\hat{\beta}) + 2k, \quad \text{BIC} = -2\ell(\hat{\beta}) + k \log n,$$

where  $k$  is the number of parameters in the model.

- Best subset selection is **computationally intensive** when  $p$  is large, to the extent that it is not even feasible in most cases, but approximations such as **forward** and **backward** selection are possible.
- In **exploratory studies**, these methods are useful if applied **cautiously**.
- As we shall discuss, an excess of automatism may lead to good predictive performance, but it may fail to make the model simple or interpretable.

## Stepwise procedures: forward and backward selection

- **Forward selection** adds terms sequentially. At each stage it selects the term giving the greatest improvement in terms of **deviance** or other goodness of fit measures.
- The process stops when further additions do not improve the fit, according to statistical significance (i.e. a log-likelihood ratio test) or a criterion for judging the model fit (such as the AIC or BIC).
- A stepwise variation of this procedure rechecks, at each stage, whether terms added at previous stages are still needed.
- **Backward elimination** begins with a complex model and sequentially removes terms.
- At each stage, it selects the term whose removal has the least damaging effect on the model, such as the largest p-value in a test or the least deterioration in a criterion for judging the model fit.
- The process stops when any further deletion leads to a poorer fit.

Whenever possible, i.e. when  $p$  is not too large, we recommend **manually** performing each stage of forward or backward procedures and **avoiding fully automatic procedures**.

# Comments on forward/backward selection I

- An **interaction term** should not be included without its **main effects**.
- For **qualitative predictors** with  $> 2$  categories: add/drop the **entire variable**, not just one indicator. Otherwise, results depend on the reference category used in coding.
- Some statisticians prefer **backward elimination** over **forward selection**. It is safer to delete terms from an overly complex model than to **add** to an overly simple one.
- Forward selection based on significance tests:
  - a. May stop prematurely if a test has **low power**.
  - b. Early-stage comparisons often involve **inadequate models**, making tests **questionable**.
- Neither backward nor forward strategies guarantee a **meaningful model**.
- Evaluating many terms increases the risk of **chance findings**. If true effects are **weak**, the largest sample effect likely **overestimates** the truth.
- Use of standard significance tests in selection lacks **theoretical justification**. The distribution of the minimum or maximum  $p$ -value at each stage is not the same as the distribution of a **pre-selected variable**. This issue is called **multiple testing** and leads to **overconfident** conclusions.

## Comments on forward/backward selection II

- Statistical significance is not the same as **practical significance**; do not rely only on significance tests.
- The price to pay for adding an irrelevant variable is an increase in **variance** of the estimates. The price to pay for dropping a relevant variable is an increase in **bias**.
- It is possible to include variables central to the study goals even if **not significant**:
  - It enables comparisons with other studies where the effect is significant, perhaps because of a larger sample size;
  - If the variable is a potential **confounder**, i.e. possibly relevant for predicting the response, but not of direct interest, including it in the model may help to reduce bias in estimating relevant effects of key explanatory variables.
- Do not keep variables just because they are **significant**.
  - As an example, consider an adjusted  $R^2 = 0.39$  in a linear model with interactions vs. 0.38 without. The simpler model may be preferable being more interpretable.
- Algorithmic selection methods are no substitute for careful thought in model building.

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